# Big Bend Seagrasses Aquatic Preserve

SEACAR Habitat Analyses

Last compiled on 10 June, 2026

## Contents

## Funding & Acknowledgements

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## Threshold Filtering

Threshold filters, following the guidance of Florida Department of Environmental Protection's (*FDEP*) Division of Environmental Assessment and Restoration (*DEAR*) are used to exclude specific results values from the SEACAR Analysis. Based on the threshold filters, Quality Assurance / Quality Control (*QAQC*) Flags are inserted into the *SEACAR_QAQCFlagCode* and *SEACAR_QAQC_Description* columns of the export data. The *Include* column indicates whether the *QAQC* Flag will also indicate that data are excluded from analysis. No data are excluded from the data export, but the analysis scripts can use the *Include* column to exclude data (1 to include, 0 to exclude).

Table 1: Continuous Water Quality threshold values

| *Parameter Name* | *Units* | *Low Threshold* | *High Threshold* |
|---|---|---|---|
| Chlorophyll a, Uncorrected for Pheophytin | ug/L | - | - |
| Dissolved Oxygen | mg/L | -0.000001 | 50 |
| Dissolved Oxygen Saturation | % | -0.000001 | 500 |
| Fluorescent Dissolved Organic Matter | QSE | - | - |
| Salinity | ppt | -0.000001 | 70 |
| Specific Conductivity | mS/cm | -0.000001 | 200 |
| Turbidity | NTU | -0.000001 | 4000 |
| Water Temperature | Degrees C | -5.000000 | 45 |
| pH | None | 2.000000 | 14 |

Table 2: Discrete Water Quality threshold values

| *Parameter Name* | *Units* | *Low Threshold* | *High Threshold* |
|---|---|---|---|
| Ammonia, Un-ionized (NH3) | mg/L | - | - |
| Ammonium (NH4) | mg/L | - | - |

| *Parameter Name* | *Units* | *Low Threshold* | *High Threshold* |
|---|---|---|---|
| Chlorophyll a, Corrected for Pheophytin | ug/L | - | - |
| Chlorophyll a, Uncorrected for Pheophytin | ug/L | - | - |
| Colored Dissolved Organic Matter | PCU | - | - |
| Dissolved Oxygen | mg/L | -0.000001 | 25 |
| Dissolved Oxygen Saturation | % | -0.000001 | 310 |
| Fluorescent Dissolved Organic Matter | QSE | - | - |
| Light Extinction Coefficient | m^-1 | - | - |
| NO2+3, Filtered | mg/L | - | - |
| Nitrate (NO3) | mg/L | - | - |
| Nitrite (NO2) | mg/L | - | - |
| Nitrogen, inorganic | mg/L | - | - |
| Nitrogen, organic | mg/L | - | - |
| Phosphate, Filtered (PO4) | mg/L | - | - |
| Salinity | ppt | -0.000001 | 70 |
| Secchi Depth | m | 0.000001 | 50 |
| Specific Conductivity | mS/cm | 0.005000 | 100 |
| Total Ammonia (N) | mg/L | - | - |
| Total Kjeldahl Nitrogen | mg/L | - | - |
| Total Nitrogen | mg/L | - | - |
| Total Nitrogen | mg/L | - | - |
| Total Phosphorus | mg/L | - | - |
| Total Suspended Solids | mg/L | - | - |
| Turbidity | NTU | - | - |
| Water Temperature | Degrees C | 3.000000 | 40 |
| pH | None | 2.000000 | 13 |

Table 3: Quality Assurance Flags inserted based on threshold checks listed in Table 1 and 2

| *SEACAR QAQC Description* | *Include* | *SEACAR QAQCFlagCode* |
|---|---|---|
| Exceeds maximum threshold | 0 | 2Q |
| Below minimum threshold | 0 | 4Q |
| Within threshold tolerance | 1 | 6Q |
| No defined thresholds for this parameter | 1 | 7Q |

## Value Qualifiers

Value qualifier codes included within the data are used to exclude certain results from the analysis. The data are retained in the data export files, but the analysis uses the *Include* column to filter the results.

### STORET and WIN value qualifier codes

Value qualifier codes from *STORET* and *WIN* data are examined with the database and used to populate the *Include* column in data exports.

Table 4: Value Qualifier codes excluded from analysis

| *Qualifier Source* | *Value Qualifier* | *Include* | *MDL* | *Description* |
|---|---|---|---|---|
| STORET-WIN | H | 0 | 0 | Value based on field kit determination; results may not be accurate |
| STORET-WIN | J | 0 | 0 | Estimated value |
| STORET-WIN | V | 0 | 0 | Analyte was detected at or above method detection limit |
| STORET-WIN | Y | 0 | 0 | Lab analysis from an improperly preserved sample; data may be inaccurate |

### Discrete Water Quality Value Qualifiers

The following value qualifiers are highlighted in the Discrete Water Quality section of this report. An exception is made for **Program 476** - *Charlotte Harbor Estuaries Volunteer Water Quality Monitoring Network* and data flagged with Value Qualifier **H** are included for this program only.

**H** - Value based on field kit determiniation; results may not be accurate. This code shall be used if a field screening test (e.g., field gas chromatograph data, immunoassay, or vendor-supplied field kit) was used to generate the value and the field kit or method has not been recognized by the Department as equivalent to laboratory methods.

**I** - The reported value is greater than or equal to the laboratory method detection limit but less than the laboratory practical quantitation limit.

**Q** - Sample held beyond the accepted holding time. This code shall be used if the value is derived from a sample that was prepared or analyzed after the approved holding time restrictions for sample preparation or analysis.

**S** - Secchi disk visible to bottom of waterbody. The value reported is the depth of the waterbody at the location of the Secchi disk measurement.

**U** - Indicates that the compound was analyzed for but not detected. This symbol shall be used to indicate that the specified component was not detected. The value associated with the qualifier shall be the laboratory method detection limit. Unless requested by the client, less than the method detection limit values shall not be reported

### Systemwide Monitoring Program (SWMP) value qualifier codes

Value qualifier codes from the *SWMP* continuous program are examined with the database and used to populate the *Include* column in data exports. *SWMP* Qualifier Codes are indicated by *QualifierSource=SWMP*.

Table 5: SWMP Value Qualifier codes

| *Qualifier Source* | *Value Qualifier* | *Include* | *Description* |
|---|---:|---:|---|
| SWMP | -1 | 1 | Optional parameter not collected |
| SWMP | -2 | 0 | Missing data |
| SWMP | -3 | 0 | Data rejected due to QA/QC |
| SWMP | -4 | 0 | Outside low sensor range |
| SWMP | -5 | 0 | Outside high sensor range |
| SWMP | 0 | 1 | Passed initial QA/QC checks |
| SWMP | 1 | 0 | Suspect data |
| SWMP | 2 | 1 | Reserved for future use |
| SWMP | 3 | 1 | Calculated data: non-vented depth/level sensor correction for changes in barometric pressure |
| SWMP | 4 | 1 | Historical: Pre-auto QA/QC |
| SWMP | 5 | 1 | Corrected data |

# Water Column

The water column habitat extends from the water's surface to the bottom sediments, and it's where fish, dolphins, crabs and people swim! So much life makes its home in the water column that the health of marine and coastal ecosystems, as well as human economies, depend on the condition of this vulnerable habitat. Local patterns of rainfall, temperature, winds and currents can rapidly change the condition of the water column, while global influences such as El Niño/La Niña, large-scale fluctuation in sea temperatures and climate change can have long-term effects. Inputs from the prosperity of our day-to-day lives including farming, mining and forestry, and emissions from power generation, automobiles and water treatment can also alter the health of the water column. Acting alone or together, each input can have complex and lasting effects on habitats and ecosystems.

SEACAR evaluates water column health with several essential parameters. These include nutrient surveys of nitrogen and phosphorus, and water quality assessments of salinity, dissolved oxygen, pH, and water temperature. Water clarity is evaluated with Secchi depth, turbidity, levels of chlorophyll a, total suspended solids, and colored dissolved organic matter. Additionally, the richness of nekton is indicated by the abundance of free-swimming fishes and macroinvertebrates like crabs and shrimps.

# Seasonal Kendall-Tau Analysis

Indicators must have a minimum of five to ten years, depending on the habitat, of data within the geographic range of the analysis to be included in the analysis. Ten years of data are required for discrete parameters, and five years of data are required for continuous parameters. If there are insufficient years of data, the number of years of data available will be noted and labeled as "insufficient data to conduct analysis". Further, for the preferred Seasonal Kendall-Tau test, there must be data from at least two months in common across at least two consecutive years within the RCP managed area being analyzed. Values that pass both of these tests will be included in the analysis and be labeled as *Use_In_Analysis* = **TRUE**. Any that fail either test will be excluded from the analyses and labeled as *Use_In_Analysis* = **FALSE**. The points for all Water Column plots displayed in this section are monthly averages. Trend significance will be denoted as "Significant Trend" (when $p < 0.05$), or "Non-significant Trend" (when $p >= 0.05$). Any parameters with insufficient data to perform Seasonal Kendall-Tau test will have their monthly averages plotted without a corresponding trend line.

# Water Quality - Discrete

The following files were used in the discrete analysis:

- *Combined_WQ_WC_NUT_Chlorophyll_a_corrected_for_pheophytin-2026-May-18.txt*
- *Combined_WQ_WC_NUT_Chlorophyll_a_uncorrected_for_pheophytin-2026-May-18.txt*
- *Combined_WQ_WC_NUT_Colored_dissolved_organic_matter_CDOM-2026-May-18.txt*
- *Combined_WQ_WC_NUT_Dissolved_Oxygen-2026-May-18.txt*
- *Combined_WQ_WC_NUT_Dissolved_Oxygen_Saturation-2026-May-18.txt*
- *Combined_WQ_WC_NUT_pH-2026-May-18.txt*
- *Combined_WQ_WC_NUT_Salinity-2026-May-18.txt*
- *Combined_WQ_WC_NUT_Secchi_Depth-2026-May-18.txt*
- *Combined_WQ_WC_NUT_Total_Nitrogen-2026-May-18.txt*
- *Combined_WQ_WC_NUT_Total_Phosphorus-2026-May-18.txt*
- *Combined_WQ_WC_NUT_Total_Suspended_Solids_TSS-2026-May-18.txt*
- *Combined_WQ_WC_NUT_Turbidity-2026-May-18.txt*
- *Combined_WQ_WC_NUT_Water_Temperature-2026-May-18.txt*

## Chlorophyll a, Corrected for Pheophytin - Discrete

### Seasonal Kendall-Tau Trend Analysis

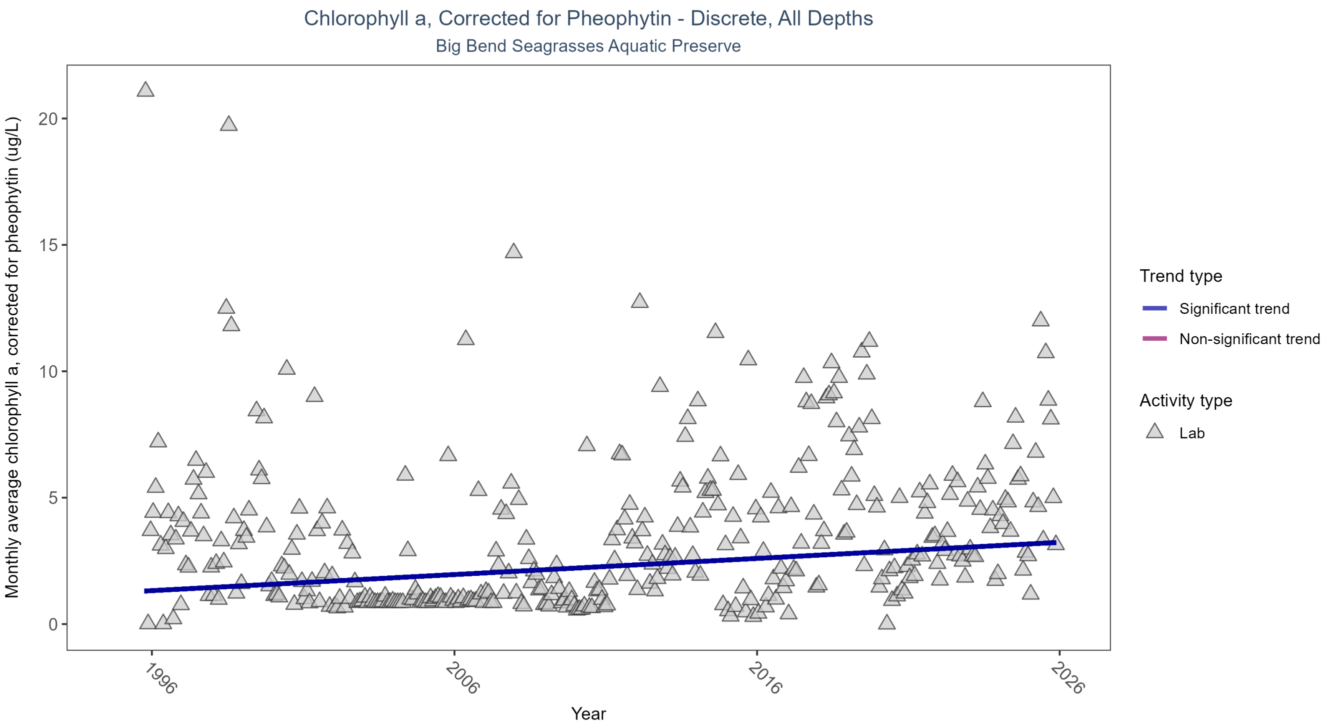


Figure 1: Scatter plot of monthly average levels of chlorophyll a, corrected for pheophytin, over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only laboratory-analyzed chlorophyll a (triangles) is included in the plot.

Table 6: Seasonal Kendall-Tau Trend Analysis for Chlorophyll a, Corrected for Pheophytin

| Activity Type | Statistical Trend | No. of Samples | No. Years with Data | Period of Record | Median Result Value | Tau | Sen Intercept | Sen Slope | P |
|---|---|---|---|---|---|---|---|---|---|
| Lab | Increasing trend | 5053 | 31 | 1995 - 2025 | 1.1 | 0.1919 | 1.2585 | 0.0637 | 0 |

Monthly average chlorophyll a, corrected for pheophytin, increased by 0.06 µg/L per year, indicating a decrease in water clarity.

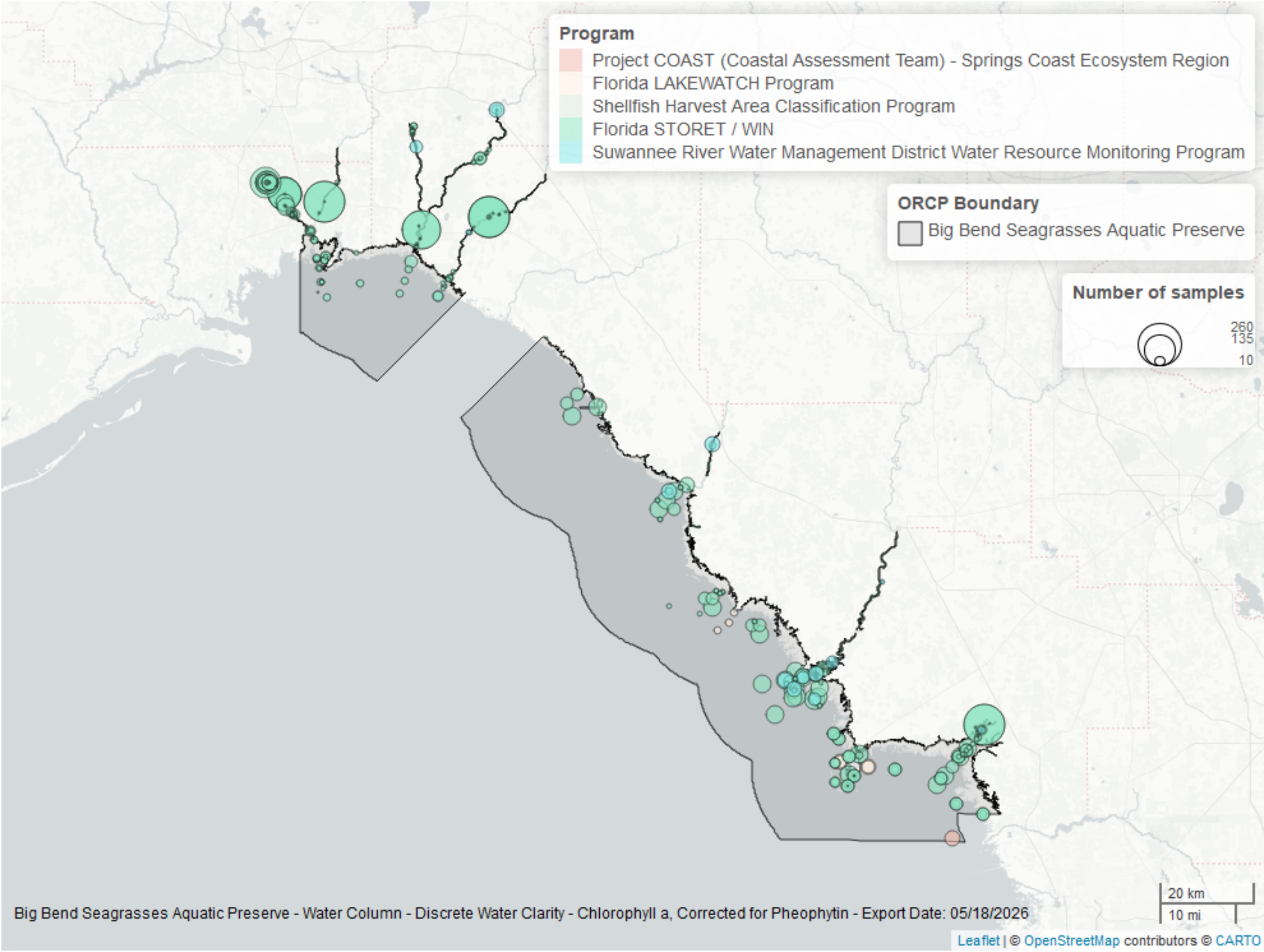


Figure 2: Map showing location of discrete water quality sampling locations within the boundaries of *Big Bend Seagrasses Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Table 7: Programs contributing data for Chlorophyll a, Corrected for Pheophytin

| ProgramID | N_Data | YearMin | YearMax |
|---|---|---|---|
| 5002 | 4262 | 1995 | 2025 |
| 514 | 378 | 2013 | 2025 |
| 477 | 338 | 2016 | 2025 |
| 540 | 131 | 2017 | 2022 |
| 5008 | 32 | 2023 | 2025 |

**Program names:**

*5002* - Florida STORET / WIN[1]
*514* - Florida LAKEWATCH Program[2]
*477* - Suwannee River Water Management District Water Resource Monitoring Program[3]
*540* - Shellfish Harvest Area Classification Program[4]
*5008* - Project COAST (Coastal Assessment Team) - Springs Coast Ecosystem Region[5]

## Chlorophyll a, Uncorrected for Pheophytin - Discrete

### Seasonal Kendall-Tau Trend Analysis

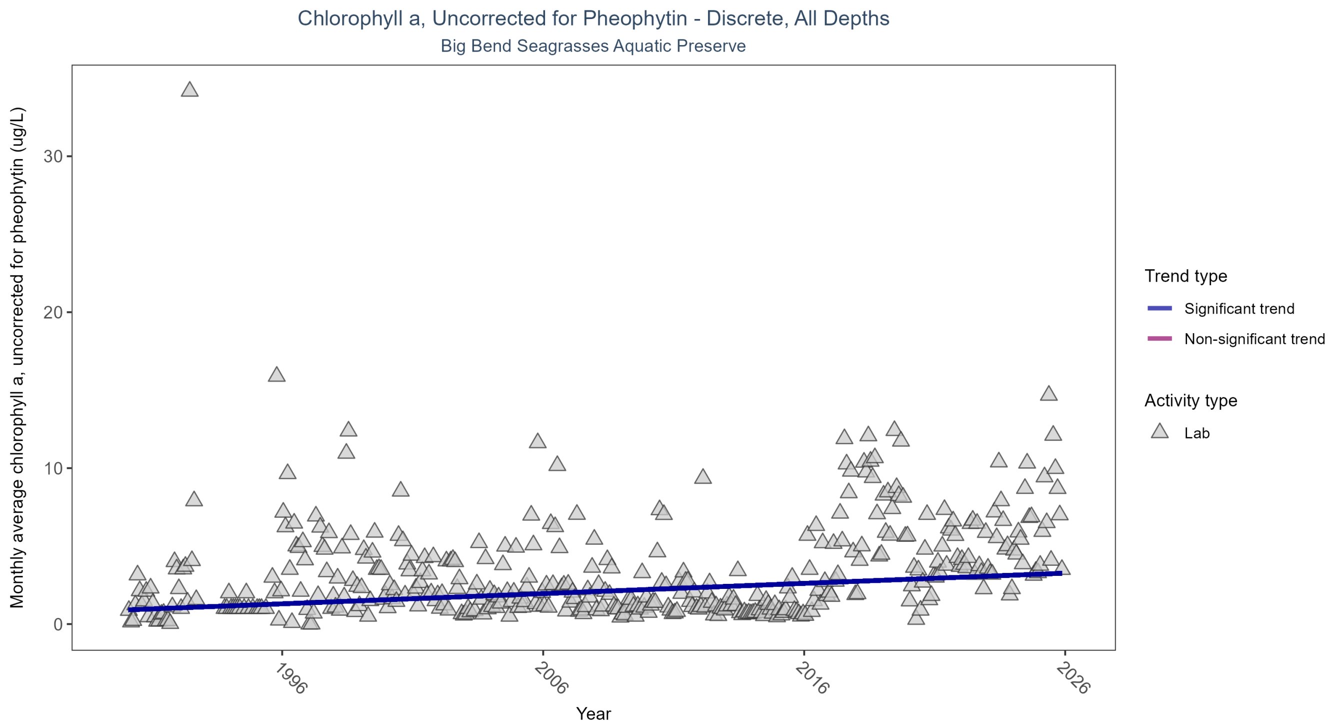


Figure 3: Scatter plot of monthly average levels of chlorophyll a, uncorrected for pheophytin, over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only laboratory-analyzed chlorophyll a (triangles) is included in the plot.

Table 8: Seasonal Kendall-Tau Trend Analysis for Chlorophyll a, Uncorrected for Pheophytin

| Activity Type | Statistical Trend | No. of Samples | No. Years with Data | Period of Record | Median Result Value | Tau | Sen Intercept | Sen Slope | P |
|---|---|---|---|---|---|---|---|---|---|
| Lab | Increasing trend | 6422 | 36 | 1990 - 2025 | 1.2 | 0.2317 | 0.9067 | 0.0655 | 0 |

Monthly average chlorophyll a, uncorrected for pheophytin, increased by 0.07 µg/L per year, indicating a decrease in water clarity.

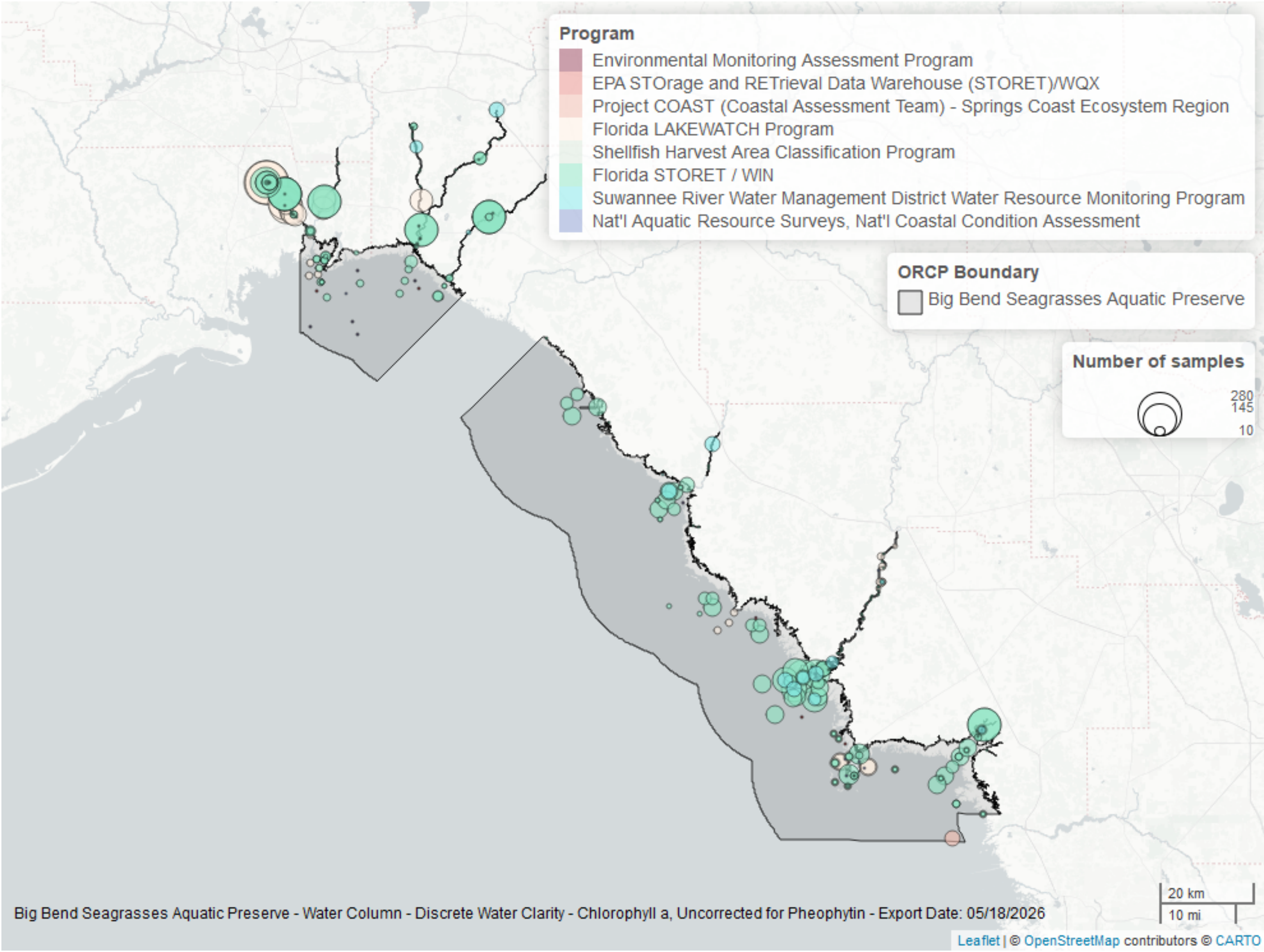


Figure 4: Map showing location of discrete water quality sampling locations within the boundaries of *Big Bend Seagrasses Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Table 9: Programs contributing data for Chlorophyll a, Uncorrected for Pheophytin

| ProgramID | N_Data | YearMin | YearMax |
|---|---|---|---|
| 5002 | 3636 | 1990 | 2025 |
| 514 | 2309 | 1993 | 2025 |
| 477 | 341 | 2016 | 2025 |
| 540 | 131 | 2017 | 2022 |
| 5008 | 33 | 2021 | 2025 |
| 103 | 22 | 2000 | 2015 |
| 118 | 12 | 2000 | 2010 |
| 115 | 7 | 2000 | 2004 |

**Program names:**

*5002* - Florida STORET / WIN[1]
*514* - Florida LAKEWATCH Program[2]
*477* - Suwannee River Water Management District Water Resource Monitoring Program[3]
*540* - Shellfish Harvest Area Classification Program[4]

*5008* - Project COAST (Coastal Assessment Team) - Springs Coast Ecosystem Region[5]
*103* - EPA STOrage and RETrieval Data Warehouse (STORET)/WQX[6]
*118* - National Aquatic Resource Surveys, National Coastal Condition Assessment[7]
*115* - Environmental Monitoring Assessment Program[8]

## Colored Dissolved Organic Matter - Discrete

### Seasonal Kendall-Tau Trend Analysis

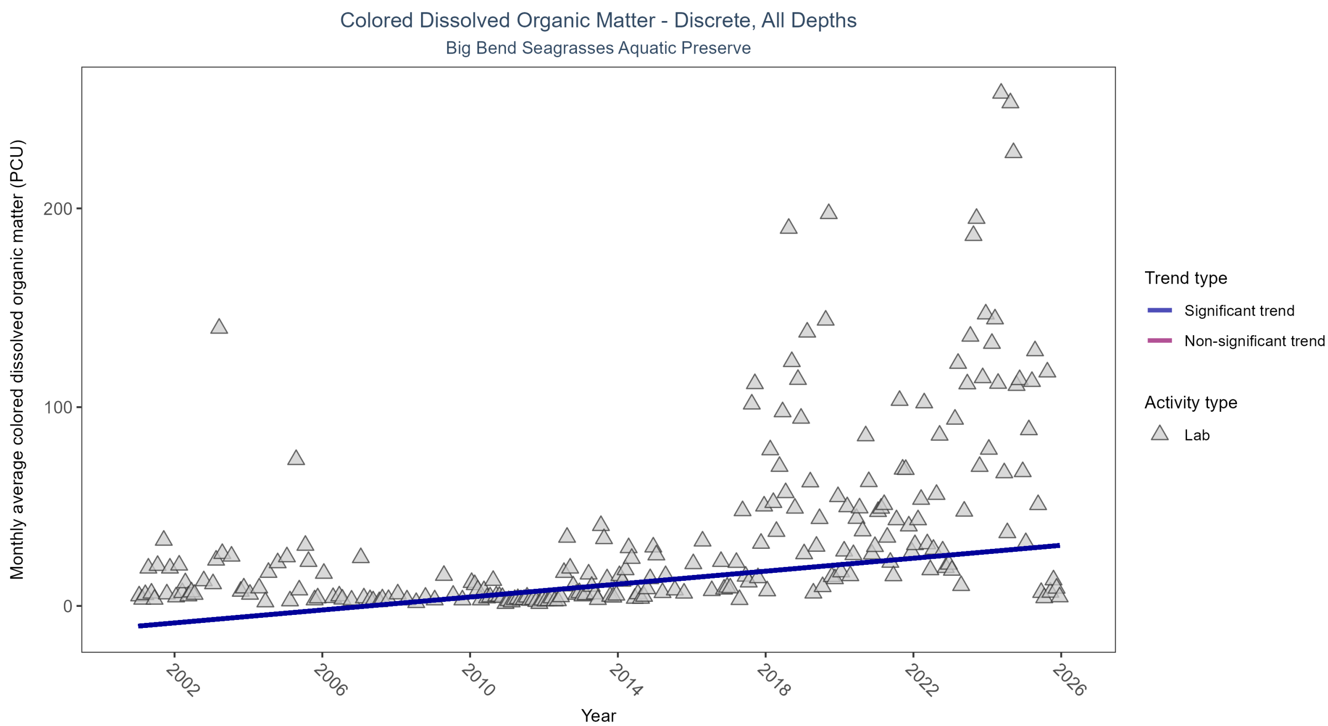


Figure 5: Scatter plot of monthly average colored dissolved organic matter (CDOM) over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only laboratory-analyzed CDOM (triangles) is included in the plot.

Table 10: Seasonal Kendall-Tau Trend Analysis for Colored Dissolved Organic Matter

| Activity Type | Statistical Trend | No. of Samples | No. Years with Data | Period of Record | Median Result Value | Tau | Sen Intercept | Sen Slope | P |
|---|---|---|---|---|---|---|---|---|---|
| Lab | Increasing trend | 2955 | 25 | 2001 - 2025 | 17.9 | 0.4498 | -10.1732 | 1.6267 | 0 |

Monthly average colored dissolved organic matter increased by 1.63 PCU per year, indicating a decrease in water clarity.

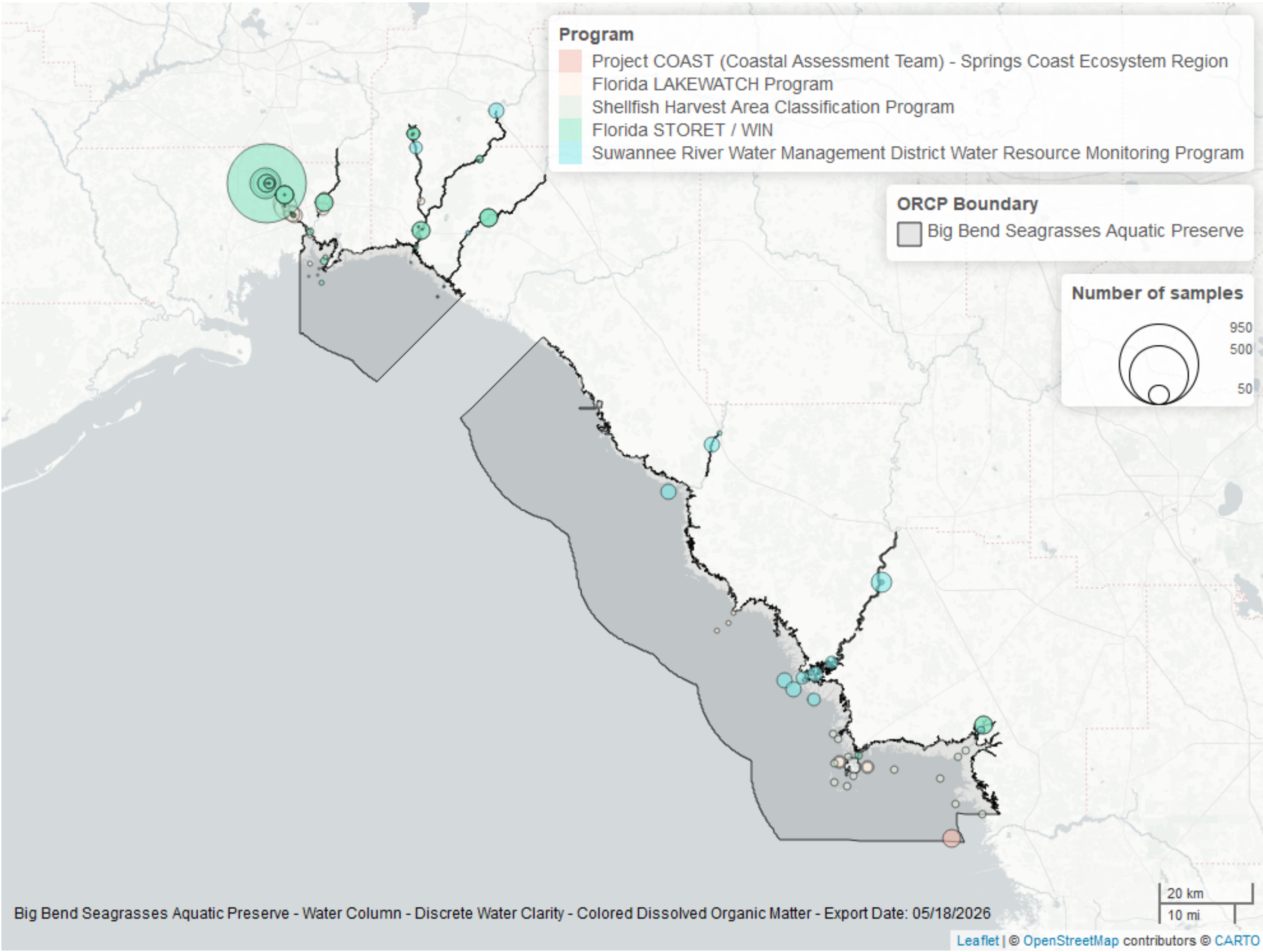


Figure 6: Map showing location of discrete water quality sampling locations within the boundaries of *Big Bend Seagrasses Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Table 11: Programs contributing data for Colored Dissolved Organic Matter

| ProgramID | N_Data | YearMin | YearMax |
|---|---|---|---|
| 5002 | 1523 | 2014 | 2025 |
| 514 | 913 | 2001 | 2025 |
| 477 | 398 | 2016 | 2025 |
| 540 | 99 | 2017 | 2019 |
| 5008 | 56 | 2021 | 2025 |

**Program names:**

*5002* - Florida STORET / WIN[1]
*514* - Florida LAKEWATCH Program[2]
*477* - Suwannee River Water Management District Water Resource Monitoring Program[3]
*540* - Shellfish Harvest Area Classification Program[4]
*5008* - Project COAST (Coastal Assessment Team) - Springs Coast Ecosystem Region[5]

## Dissolved Oxygen - Discrete

### Seasonal Kendall-Tau Trend Analysis

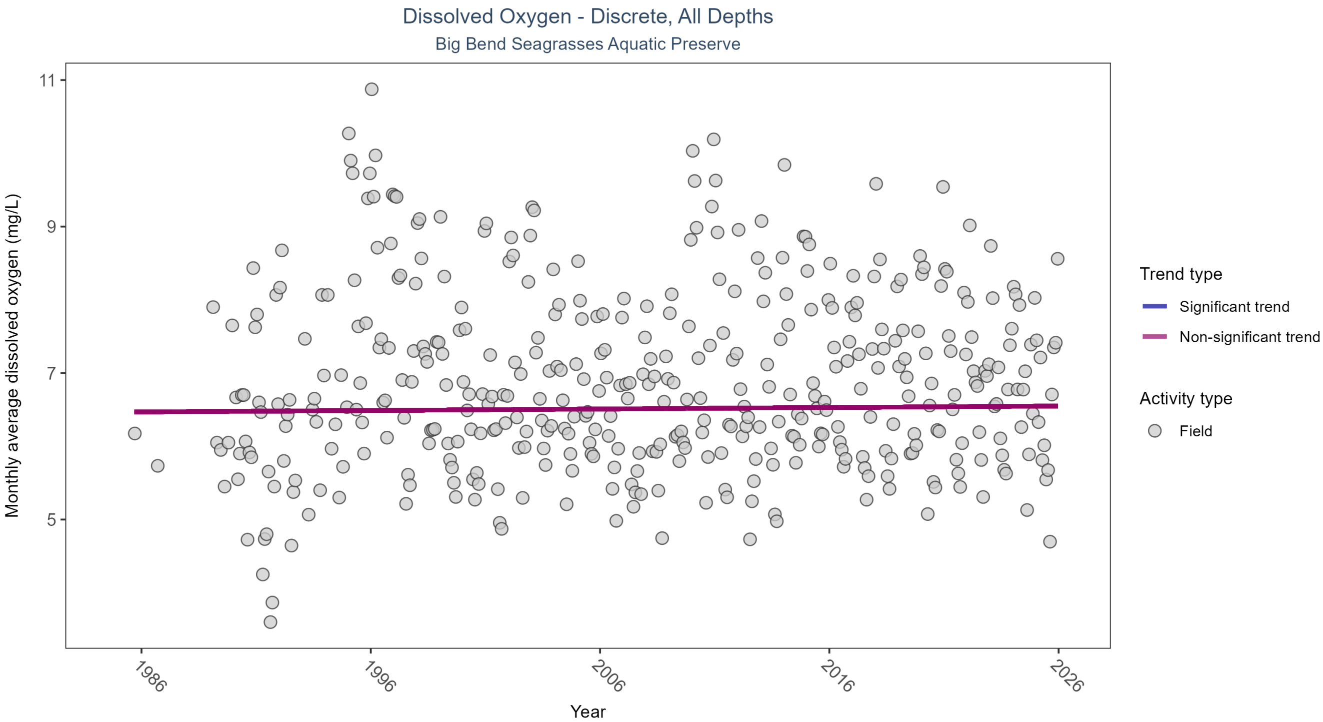


Figure 7: Scatter plot of monthly average dissolved oxygen over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only dissolved oxygen values measured in the field (circles) are included in the plot.

Table 12: Seasonal Kendall-Tau Trend Analysis for Dissolved Oxygen

| Activity Type | Statistical Trend | No. of Samples | No. Years with Data | Period of Record | Median Result Value | Tau | Sen Intercept | Sen Slope | P |
|---|---|---|---|---|---|---|---|---|---|
| Field | No detectable trend | 153302 | 39 | 1985 - 2025 | 6.7 | 0.0145 | 6.4667 | 0.002 | 0.632 |

Dissolved oxygen showed no detectable trend between 1985 and 2025.

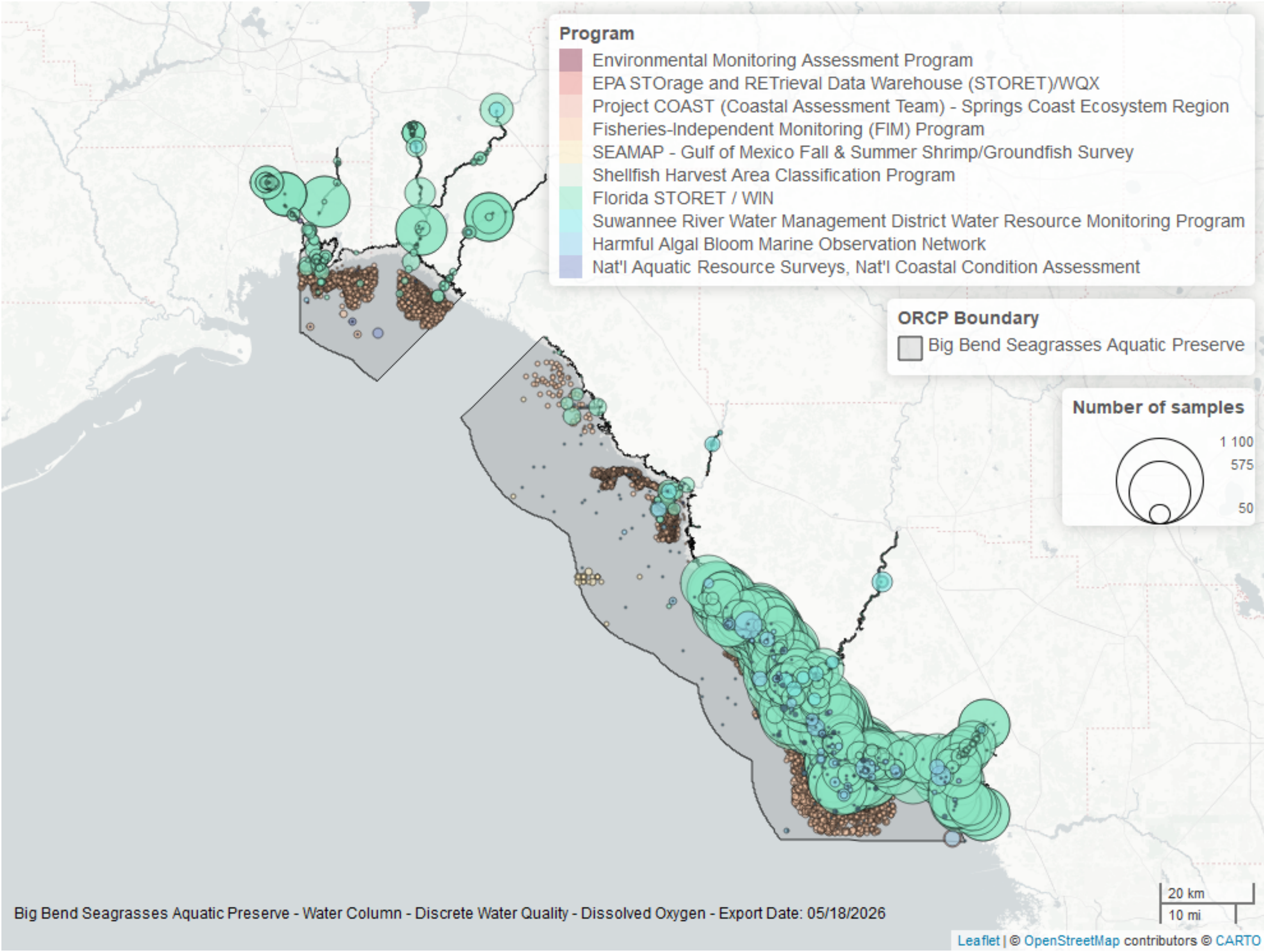


Figure 8: Map showing location of discrete water quality sampling locations within the boundaries of *Big Bend Seagrasses Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Table 13: Programs contributing data for Dissolved Oxygen

| ProgramID | N_Data | YearMin | YearMax |
|---|---|---|---|
| 5002 | 94210 | 1989 | 2025 |
| 69 | 57270 | 1996 | 2024 |
| 95 | 1161 | 1985 | 2018 |
| 477 | 398 | 2016 | 2025 |
| 540 | 121 | 2017 | 2022 |
| 60 | 113 | 1986 | 2014 |
| 5008 | 56 | 2021 | 2025 |
| 118 | 53 | 2000 | 2021 |
| 115 | 43 | 1991 | 2004 |
| 103 | 8 | 2015 | 2015 |

**Program names:**

*5002* - Florida STORET / WIN[1]
*69* - Fisheries-Independent Monitoring (FIM) Program[9]

*95* - Harmful Algal Bloom Marine Observation Network[10]
*477* - Suwannee River Water Management District Water Resource Monitoring Program[3]
*540* - Shellfish Harvest Area Classification Program[4]
*60* - Southeast Area Monitoring and Assessment Program (SEAMAP) - Gulf of Mexico Fall & Summer Shrimp/Groundfish Survey[11]
*5008* - Project COAST (Coastal Assessment Team) - Springs Coast Ecosystem Region[5]
*118* - National Aquatic Resource Surveys, National Coastal Condition Assessment[7]
*115* - Environmental Monitoring Assessment Program[8]
*103* - EPA STOrage and RETrieval Data Warehouse (STORET)/WQX[6]

## Dissolved Oxygen Saturation - Discrete

### Seasonal Kendall-Tau Trend Analysis

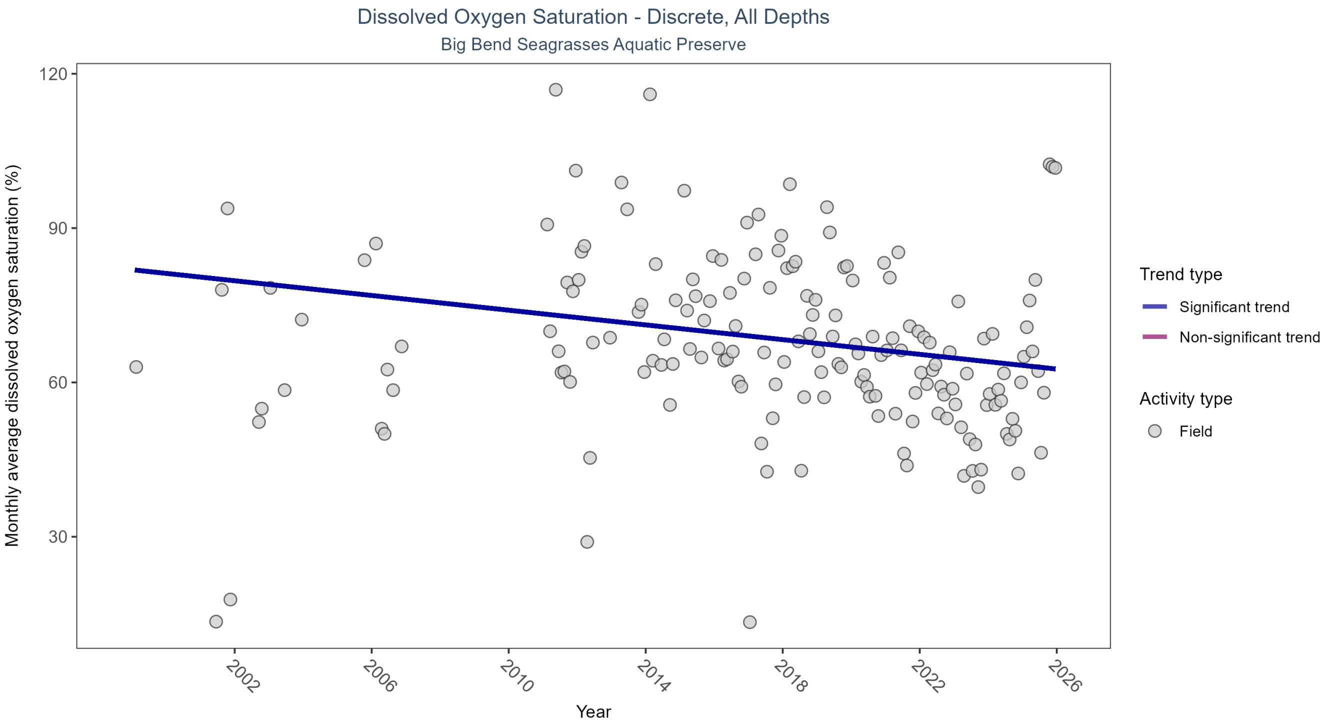


Figure 9: Scatter plot of monthly average dissolved oxygen saturation over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only dissolved oxygen saturation values measured in the field (circles) are included in the plot.

Table 14: Seasonal Kendall-Tau Trend Analysis for Dissolved Oxygen Saturation

| Activity Type | Statistical Trend | No. of Samples | No. Years with Data | Period of Record | Median Result Value | Tau | Sen Intercept | Sen Slope | P |
|---|---|---|---|---|---|---|---|---|---|
| Field | Decreasing trend | 1987 | 21 | 1999 - 2025 | 67.1 | -0.1875 | 81.9092 | -0.7152 | 0.0018 |

Monthly average dissolved oxygen saturation decreased by 0.72% per year.

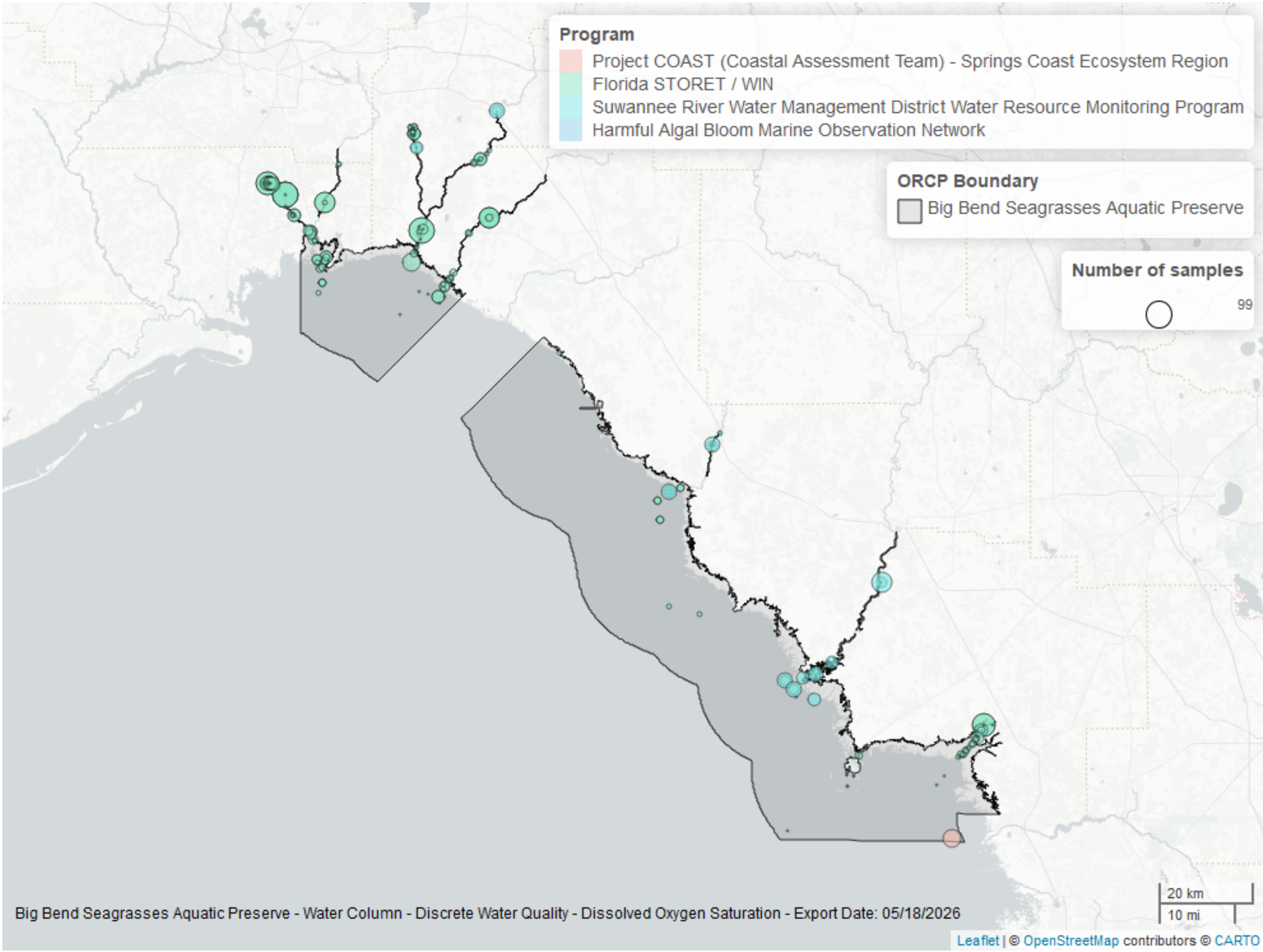


Figure 10: Map showing location of discrete water quality sampling locations within the boundaries of *Big Bend Seagrasses Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Table 15: Programs contributing data for Dissolved Oxygen Saturation

| ProgramID | N_Data | YearMin | YearMax |
|---|---|---|---|
| 5002 | 1570 | 1999 | 2025 |
| 477 | 398 | 2016 | 2025 |
| 5008 | 52 | 2021 | 2025 |
| 95 | 3 | 2016 | 2018 |

**Program names:**

*5002* - Florida STORET / WIN[1]
*477* - Suwannee River Water Management District Water Resource Monitoring Program[3]
*5008* - Project COAST (Coastal Assessment Team) - Springs Coast Ecosystem Region[5]
*95* - Harmful Algal Bloom Marine Observation Network[10]

## pH - Discrete

### Seasonal Kendall-Tau Trend Analysis

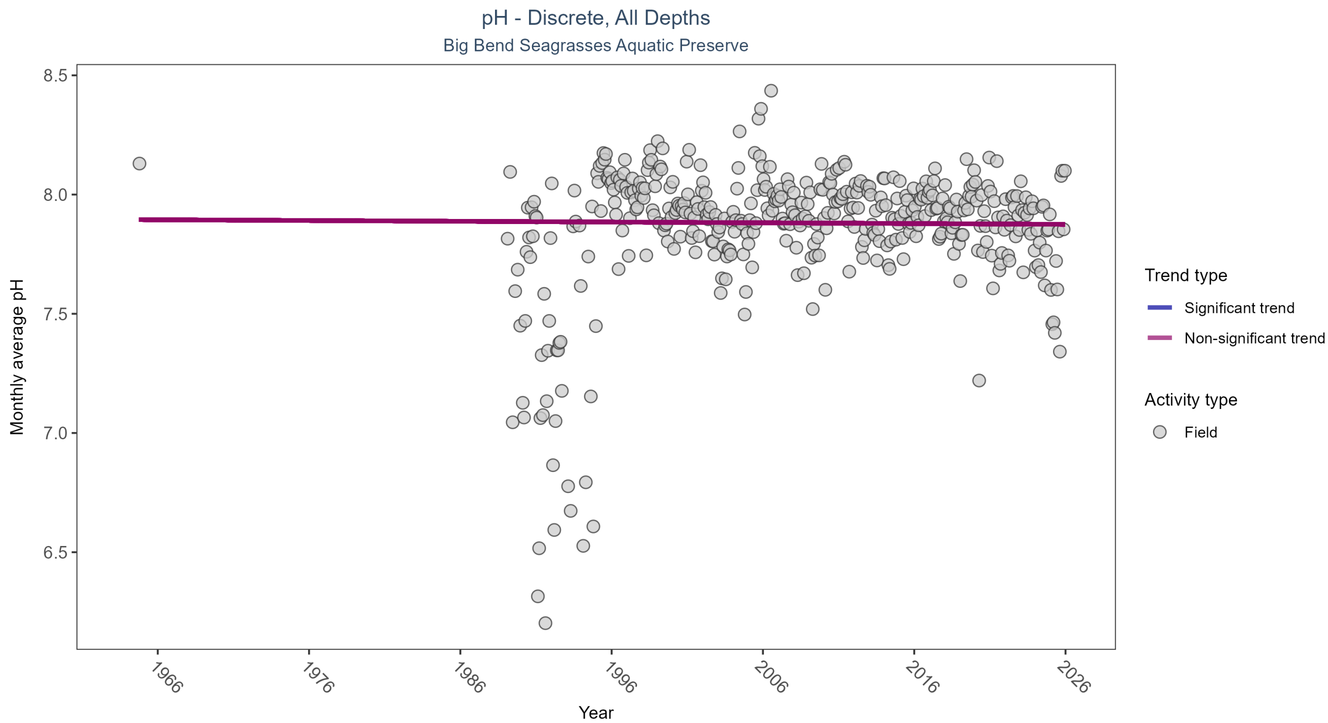


Figure 11: Scatter plot of monthly average pH over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only pH values measured in the field (circles) are included in the plot.

Table 16: Seasonal Kendall-Tau Trend Analysis for pH

| Activity Type | Statistical Trend | No. of Samples | No. Years with Data | Period of Record | Median Result Value | Tau | Sen Intercept | Sen Slope | P |
|---|---|---|---|---|---|---|---|---|---|
| Field | No detectable trend | 110666 | 38 | 1964 - 2025 | 8 | -0.0136 | 7.895 | -0.0003 | 0.7846 |

pH showed no detectable trend between 1964 and 2025.

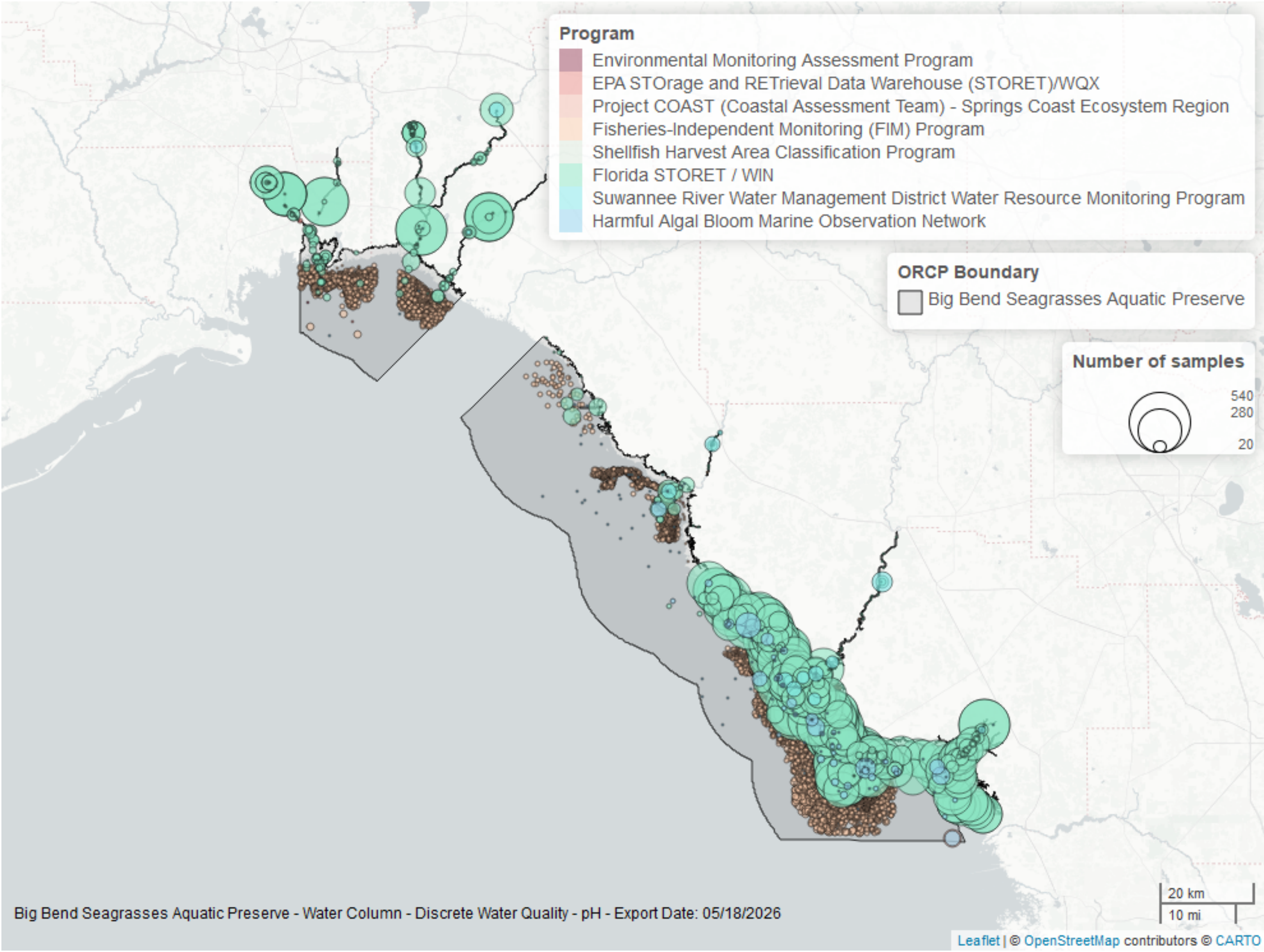


Figure 12: Map showing location of discrete water quality sampling locations within the boundaries of *Big Bend Seagrasses Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Table 17: Programs contributing data for pH

| ProgramID | N_Data | YearMin | YearMax |
|---|---|---|---|
| 69 | 57163 | 1996 | 2024 |
| 5002 | 52226 | 1989 | 2025 |
| 95 | 774 | 1964 | 2018 |
| 477 | 398 | 2016 | 2025 |
| 540 | 85 | 2017 | 2022 |
| 5008 | 53 | 2021 | 2025 |
| 115 | 43 | 1991 | 2004 |
| 103 | 8 | 2015 | 2015 |

**Program names:**

*69* - Fisheries-Independent Monitoring (FIM) Program[9]
*5002* - Florida STORET / WIN[1]
*95* - Harmful Algal Bloom Marine Observation Network[10]
*477* - Suwannee River Water Management District Water Resource Monitoring Program[3]

*540* - Shellfish Harvest Area Classification Program[4]
*5008* - Project COAST (Coastal Assessment Team) - Springs Coast Ecosystem Region[5]
*115* - Environmental Monitoring Assessment Program[8]
*103* - EPA STOrage and RETrieval Data Warehouse (STORET)/WQX[6]

## Salinity - Discrete

### Seasonal Kendall-Tau Trend Analysis

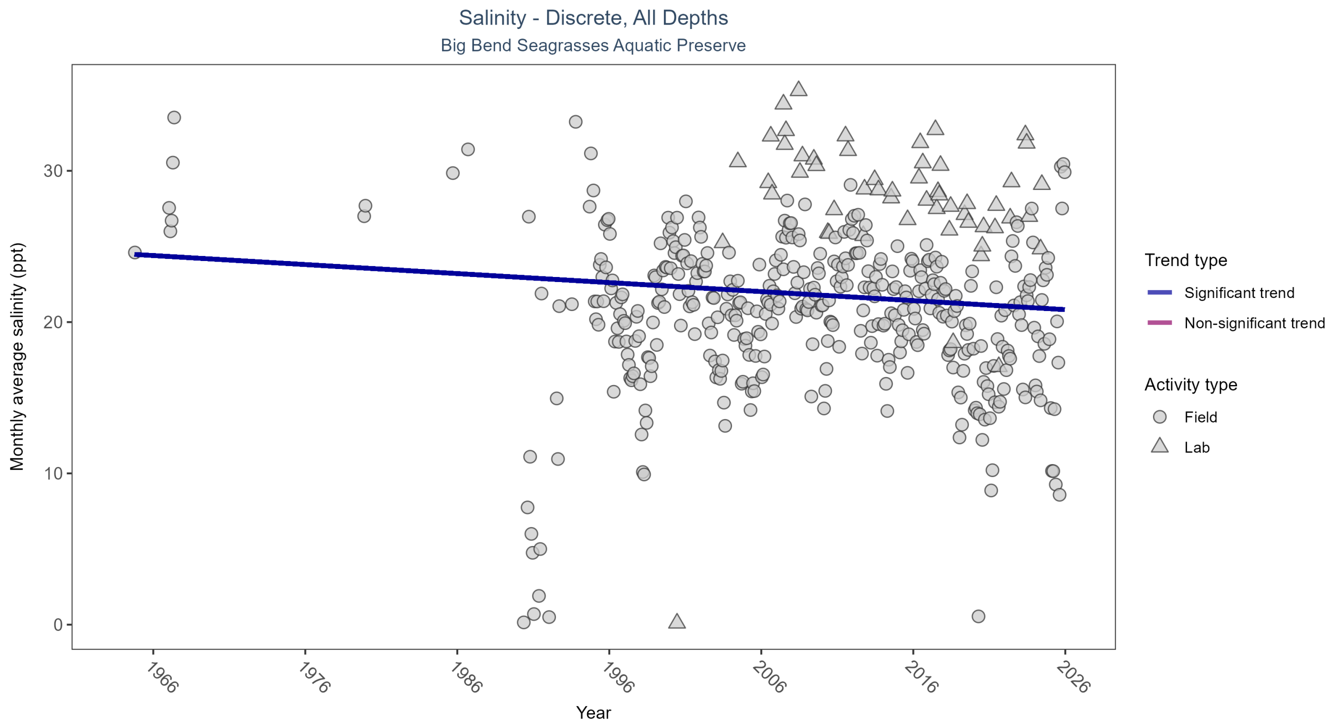


Figure 13: Scatter plot of monthly average salinity over time. If the time series included ten or more years of discrete observations, significant (blue) or non-significant (magenta) trend lines are also shown. Discrete salinity values derived from grab samples analyzed in the field (circles) or the laboratory (triangles) are both included in the plot.

Table 18: Seasonal Kendall-Tau Trend Analysis for Salinity

| Activity Type | Statistical Trend | No. of Samples | No. Years with Data | Period of Record | Median Result Value | Tau | Sen Intercept | Sen Slope | P |
|---|---|---|---|---|---|---|---|---|---|
| All | Decreasing trend | 158389 | 41 | 1964 - 2025 | 23.3 | -0.102 | 24.5188 | -0.0596 | 0.004 |

Monthly average salinity decreased by 0.06 ppt per year.

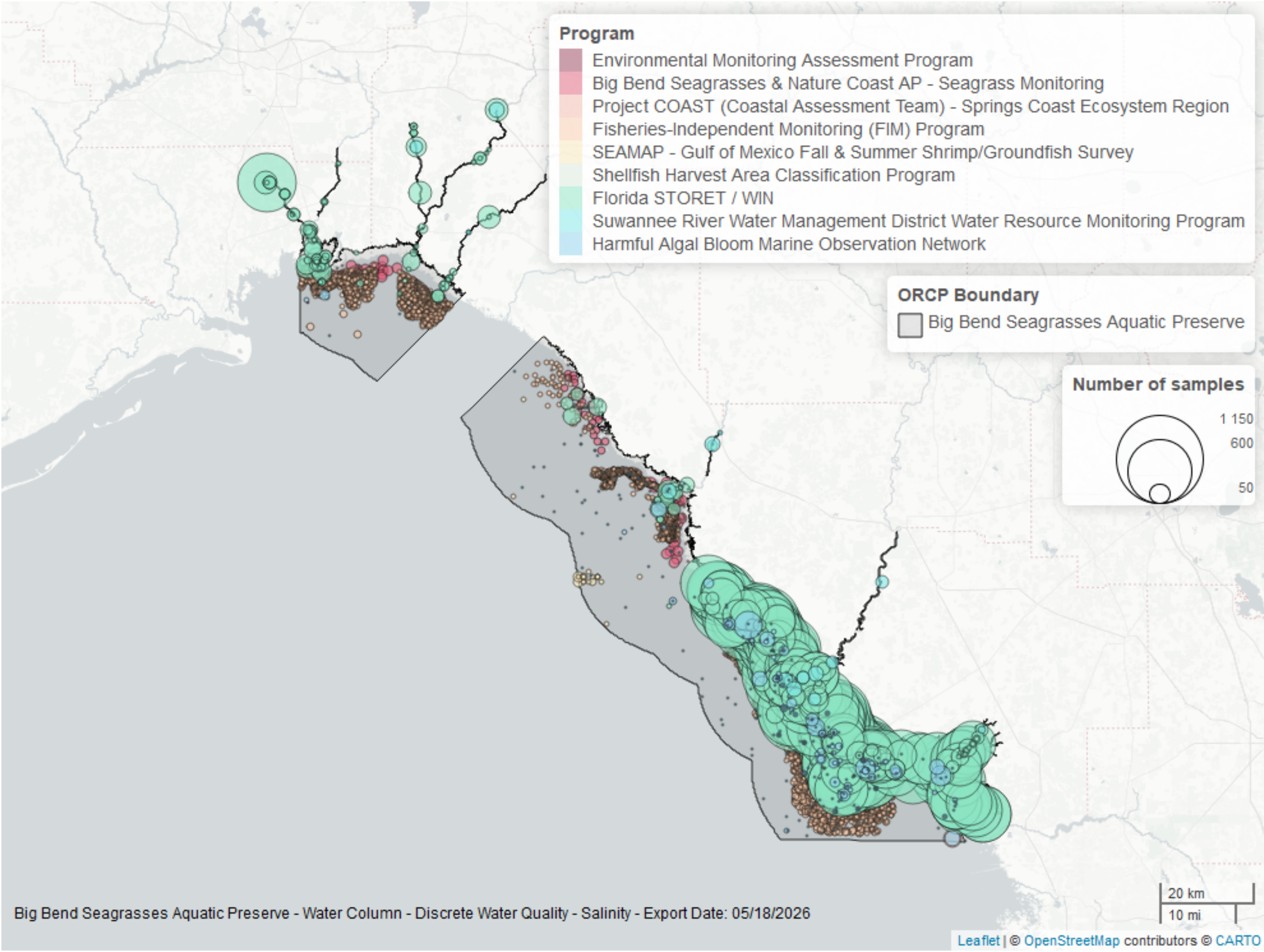


Figure 14: Map showing location of discrete water quality sampling locations within the boundaries of *Big Bend Seagrasses Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Table 19: Programs contributing data for Salinity

| ProgramID | N_Data | YearMin | YearMax |
|---|---|---|---|
| 5002 | 97584 | 1990 | 2025 |
| 69 | 57423 | 1996 | 2024 |
| 560 | 1474 | 2003 | 2024 |
| 95 | 1287 | 1964 | 2018 |
| 477 | 369 | 2016 | 2025 |
| 540 | 132 | 2017 | 2022 |
| 60 | 109 | 1986 | 2014 |
| 5008 | 55 | 2021 | 2025 |
| 115 | 43 | 1991 | 2004 |

**Program names:**

*5002* - Florida STORET / WIN[1]
*69* - Fisheries-Independent Monitoring (FIM) Program[9]
*560* - Big Bend Seagrasses & Nature Coast Aquatic Preserves - Seagrass Monitoring[12]

*95* - Harmful Algal Bloom Marine Observation Network[10]
*477* - Suwannee River Water Management District Water Resource Monitoring Program[3]
*540* - Shellfish Harvest Area Classification Program[4]
*60* - Southeast Area Monitoring and Assessment Program (SEAMAP) - Gulf of Mexico Fall & Summer Shrimp/Groundfish Survey[11]
*5008* - Project COAST (Coastal Assessment Team) - Springs Coast Ecosystem Region[5]
*115* - Environmental Monitoring Assessment Program[8]

## Total Nitrogen - Discrete

### Total Nitrogen Calculation:

The logic for calculated Total Nitrogen was provided by Kevin O'Donnell and colleagues at FDEP (with the help of Jay Silvanima, Watershed Monitoring Section). The following logic is used, in this order, based on the availability of specific nitrogen components.

1) TN = TKN + NO3O2;
2) TN = TKN + NO3 + NO2;
3) TN = ORGN + NH4 + NO3O2;
4) TN = ORGN + NH4 + NO2 + NO3;
5) TN = TKN + NO3;
6) TN = ORGN + NH4 + NO3;

Additional Information:

- Rules for use of sample fraction:
  - Florida Department of Environmental Protection (FDEP) report that if both "Total" and "Dissolved" components are reported, only "Total" is used. If the total is not reported, then the dissolved components are used as a best available replacement.
  - Total nitrogen calculations are done using nitrogen components with the same sample fraction, nitrogen components with mixed total/dissolved sample fractions are not used. In other words, total nitrogen can be calculated when TKN and NO3O2 are both total sample fractions, or when both are dissolved sample fractions. *Future calculations of total nitrogen values may be based on components with mixed sample fractions.*
- Values inserted into data:
  - ParameterName = "Total Nitrogen"
  - SEACAR_QAQCFlagCode = "1Q"
  - SEACAR_QAQC_Description = "SEACAR Calculated"

### Seasonal Kendall-Tau Trend Analysis

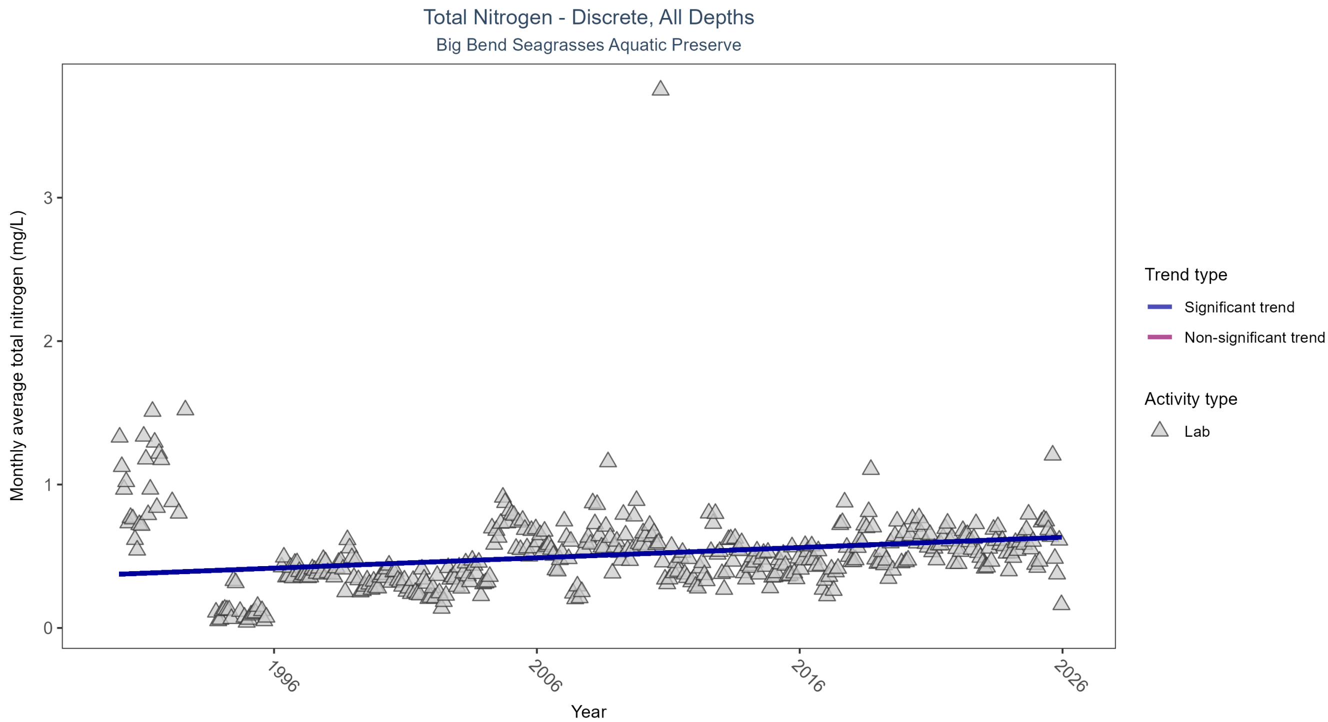


Figure 15: Scatter plot of monthly average total nitrogen over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only nitrogen values obtained from laboratory analyses (triangles) are included in the plot.

Table 20: Seasonal Kendall-Tau Trend Analysis for Total Nitrogen

| Activity Type | Statistical Trend | No. of Samples | No. Years with Data | Period of Record | Median Result Value | Tau | Sen Intercept | Sen Slope | P |
|---|---|---|---|---|---|---|---|---|---|
| Lab | Increasing trend | 8320 | 36 | 1990 - 2025 | 0.466 | 0.2279 | 0.3741 | 0.0072 | 0 |

Monthly average total nitrogen increased by 0.01 mg/L per year.

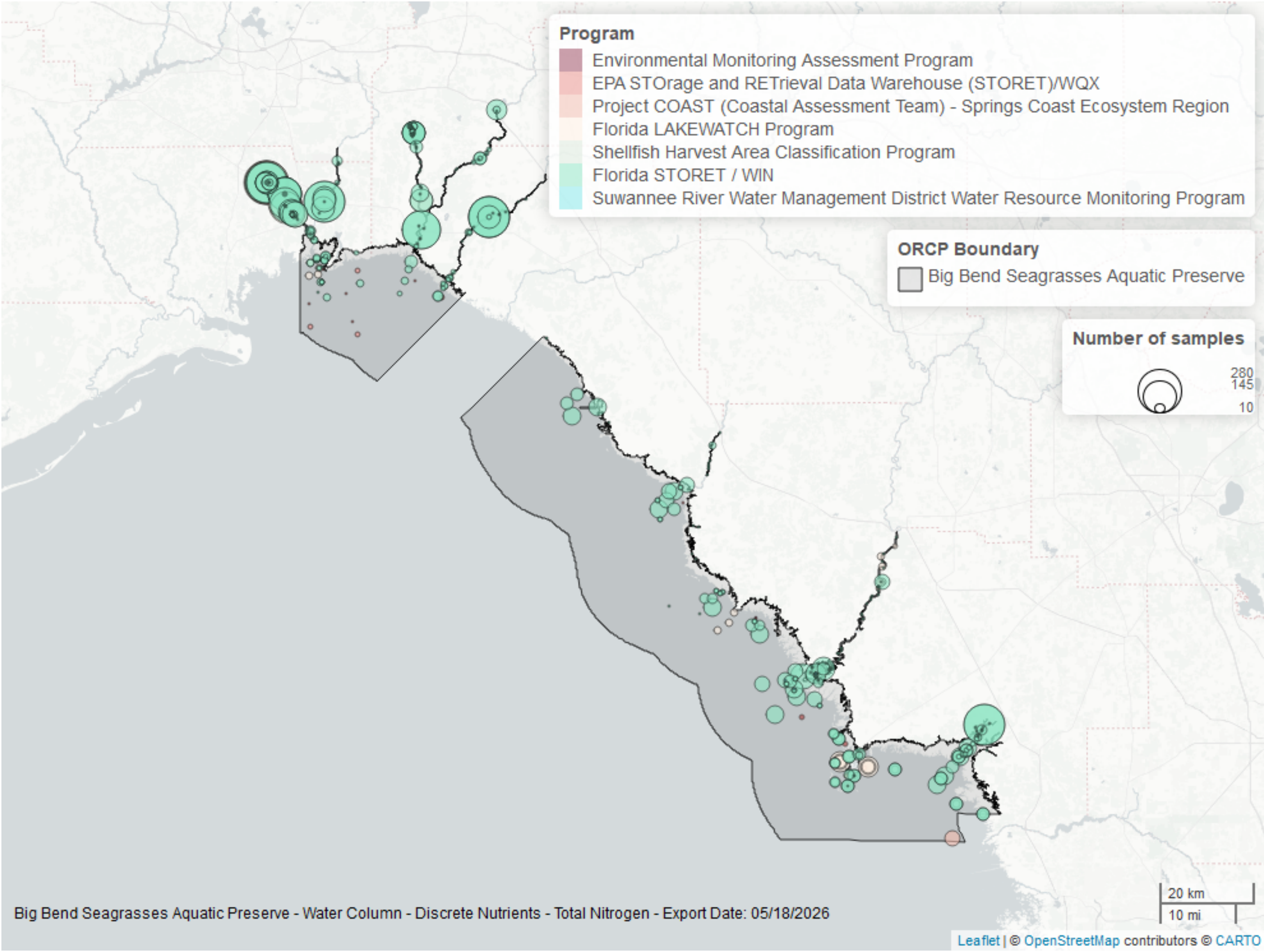


Figure 16: Map showing location of discrete water quality sampling locations within the boundaries of *Big Bend Seagrasses Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Table 21: Programs contributing data for Total Nitrogen

| ProgramID | N_Data | YearMin | YearMax |
|---|---|---|---|
| 5002 | 5684 | 1990 | 2025 |
| 514 | 2424 | 1993 | 2025 |
| 540 | 131 | 2017 | 2022 |
| 103 | 43 | 2000 | 2015 |
| 5008 | 34 | 2021 | 2025 |
| 115 | 5 | 2000 | 2004 |
| 477 | 3 | 2017 | 2017 |

**Program names:**

*5002* - Florida STORET / WIN[1]
*514* - Florida LAKEWATCH Program[2]
*540* - Shellfish Harvest Area Classification Program[4]
*103* - EPA STOrage and RETrieval Data Warehouse (STORET)/WQX[6]
*5008* - Project COAST (Coastal Assessment Team) - Springs Coast Ecosystem Region[5]

*115* - Environmental Monitoring Assessment Program[8]
*477* - Suwannee River Water Management District Water Resource Monitoring Program[3]

## Total Phosphorus - Discrete

### Seasonal Kendall-Tau Trend Analysis

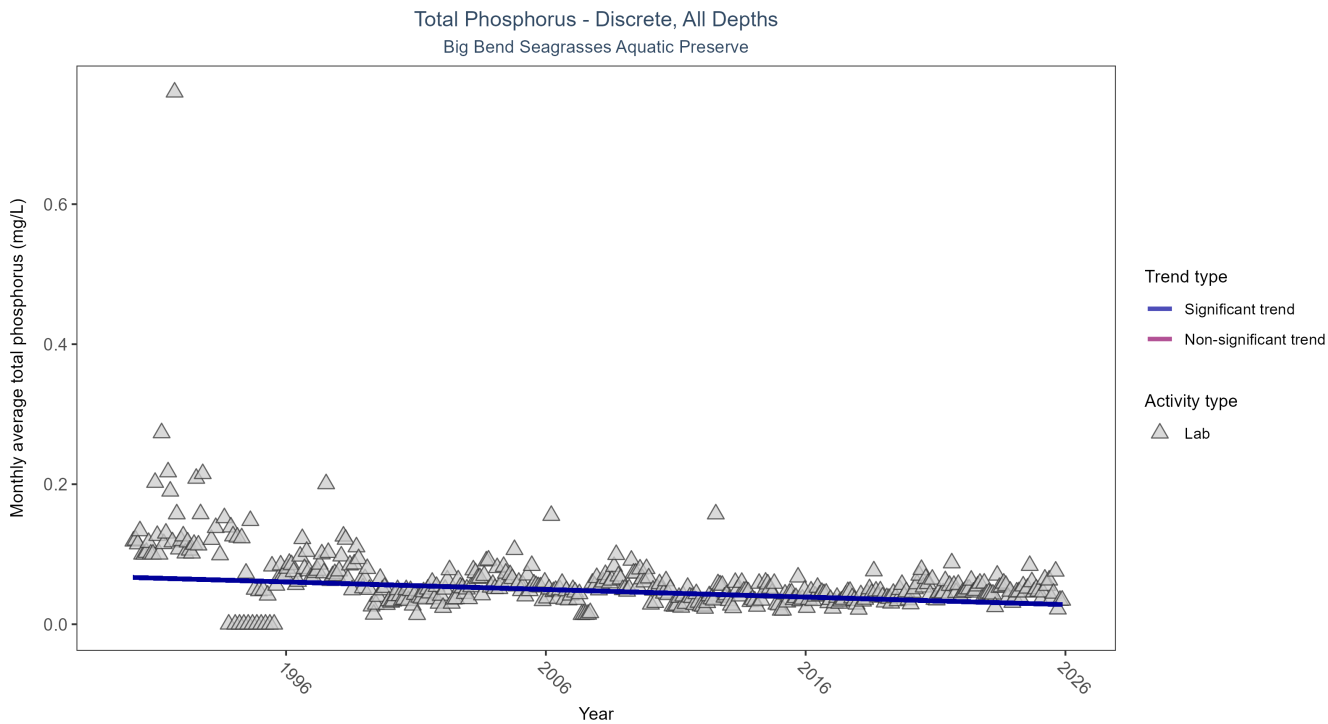


Figure 17: Scatter plot of monthly average total phosphorus over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only phosphorus values obtained from laboratory analyses (triangles) are included in the plot.

Table 22: Seasonal Kendall-Tau Trend Analysis for Total Phosphorus

| Activity Type | Statistical Trend | No. of Samples | No. Years with Data | Period of Record | Median Result Value | Tau | Sen Intercept | Sen Slope | P |
|---|---|---|---|---|---|---|---|---|---|
| Lab | Decreasing trend | 8781 | 36 | 1990 - 2025 | 0.042 | -0.2885 | 0.0669 | -0.0011 | 0 |

Monthly average total phosphorus decreased by less than 0.01 mg/L per year.

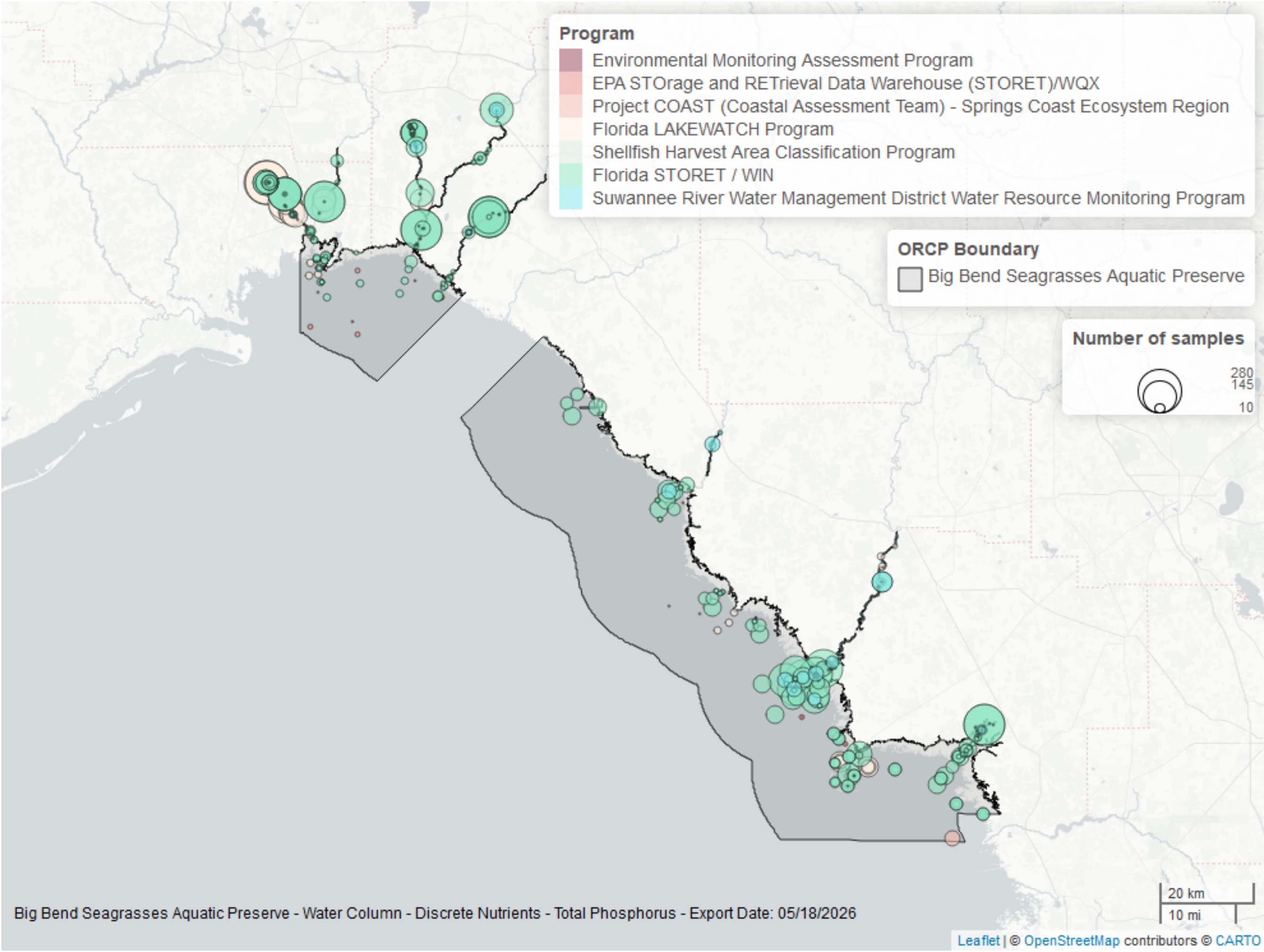


Figure 18: Map showing location of discrete water quality sampling locations within the boundaries of *Big Bend Seagrasses Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Table 23: Programs contributing data for Total Phosphorus

| ProgramID | N_Data | YearMin | YearMax |
|---|---|---|---|
| 5002 | 5933 | 1990 | 2025 |
| 514 | 2419 | 1993 | 2025 |
| 477 | 398 | 2016 | 2025 |
| 540 | 131 | 2017 | 2022 |
| 103 | 37 | 2000 | 2015 |
| 5008 | 33 | 2021 | 2025 |
| 115 | 5 | 2000 | 2004 |

**Program names:**

*5002* - Florida STORET / WIN[1]
*514* - Florida LAKEWATCH Program[2]
*477* - Suwannee River Water Management District Water Resource Monitoring Program[3]
*540* - Shellfish Harvest Area Classification Program[4]
*103* - EPA STOrage and RETrieval Data Warehouse (STORET)/WQX[6]

*5008* - Project COAST (Coastal Assessment Team) - Springs Coast Ecosystem Region[5]
*115* - Environmental Monitoring Assessment Program[8]

## Turbidity - Discrete

### Seasonal Kendall-Tau Trend Analysis

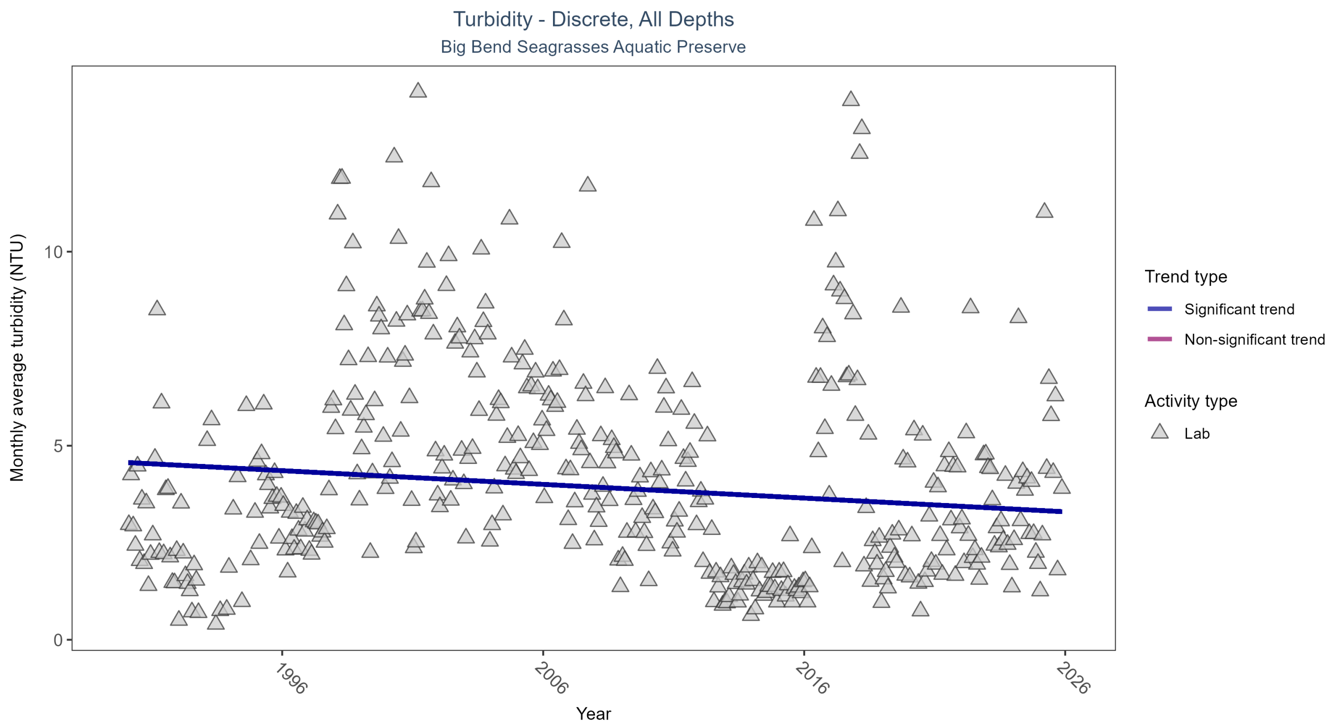


Figure 19: Scatter plot of monthly average turbidity over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only turbidity values measured in the laboratory (triangles) are included in the plot.

Table 24: Seasonal Kendall-Tau Trend Analysis for Turbidity

| Activity Type | Statistical Trend | No. of Samples | No. Years with Data | Period of Record | Median Result Value | Tau | Sen Intercept | Sen Slope | P |
|---|---|---|---|---|---|---|---|---|---|
| Lab | Decreasing trend | 43028 | 36 | 1990 - 2025 | 3.4 | -0.1063 | 4.5694 | -0.0353 | 0.002 |

Monthly average turbidity decreased by 0.04 NTU per year, indicating an increase in water clarity.

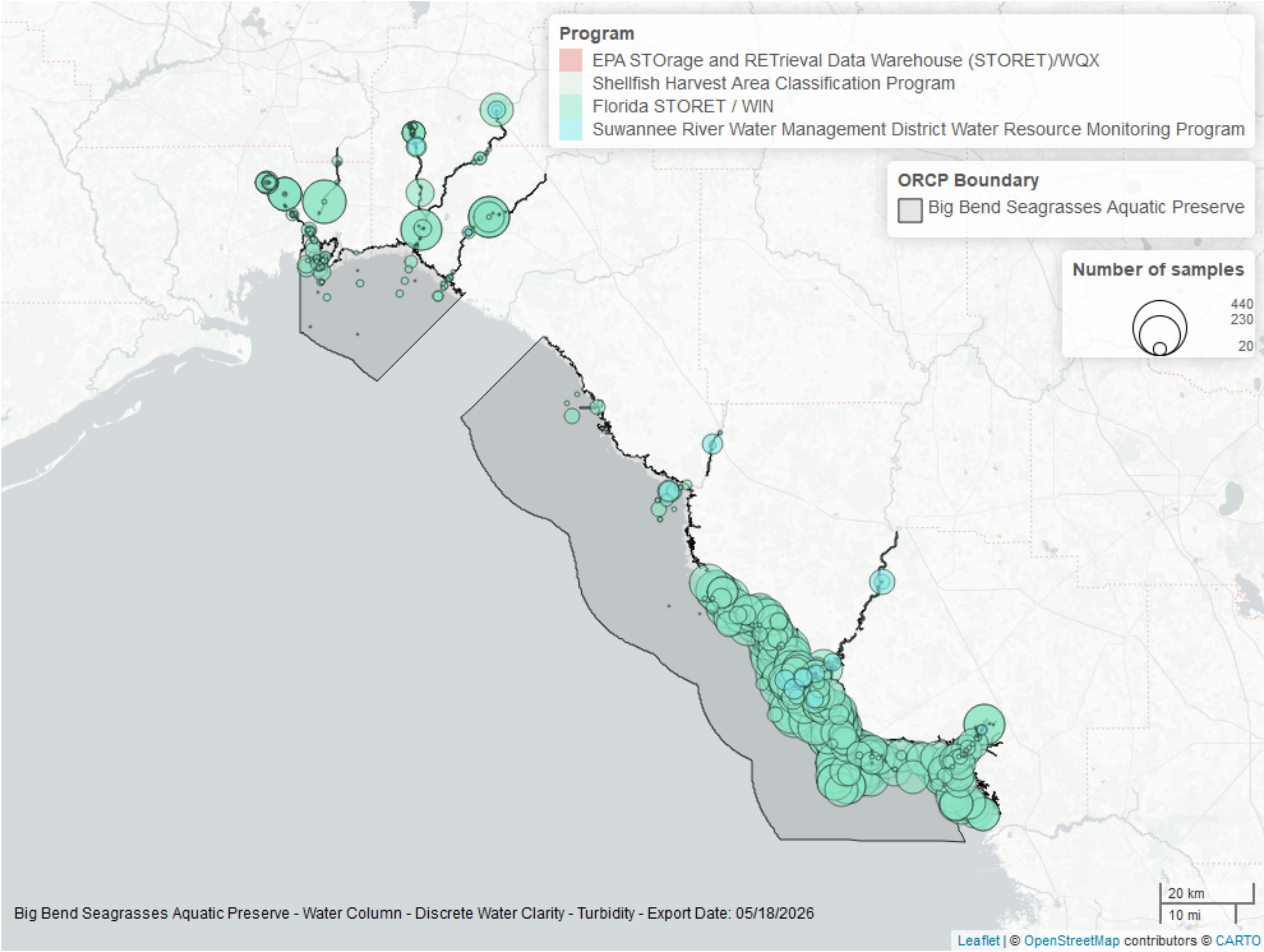


Figure 20: Map showing location of discrete water quality sampling locations within the boundaries of *Big Bend Seagrasses Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Table 25: Programs contributing data for Turbidity

| ProgramID | N_Data | YearMin | YearMax |
|---|---|---|---|
| 5002 | 42755 | 1990 | 2025 |
| 477 | 704 | 2016 | 2025 |
| 540 | 35 | 2019 | 2022 |
| 103 | 11 | 2005 | 2006 |

**Program names:**

*5002* - Florida STORET / WIN[1]
*477* - Suwannee River Water Management District Water Resource Monitoring Program[3]
*540* - Shellfish Harvest Area Classification Program[4]
*103* - EPA STOrage and RETrieval Data Warehouse (STORET)/WQX[6]

## Water Temperature - Discrete

### Seasonal Kendall-Tau Trend Analysis

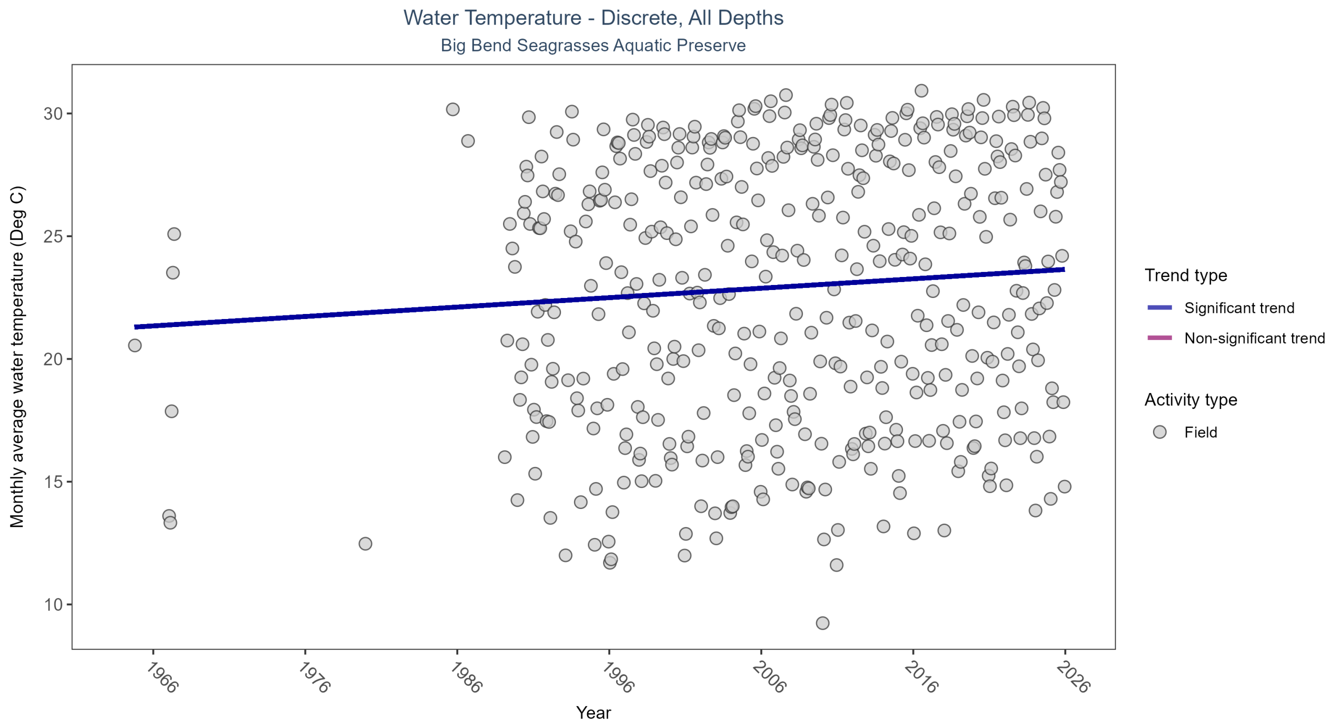


Figure 21: Scatter plot of monthly average water temperature over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only water temperature measurements taken in the field (circles) are included in the plot.

Table 26: Seasonal Kendall-Tau Trend Analysis for Water Temperature

| Activity Type | Statistical Trend | No. of Samples | No. Years with Data | Period of Record | Median Result Value | Tau | Sen Intercept | Sen Slope | P |
|---|---|---|---|---|---|---|---|---|---|
| Field | Increasing trend | 160088 | 42 | 1964 - 2025 | 24.1 | 0.2101 | 21.269 | 0.0384 | 0 |

Monthly average water temperature increased by 0.04°C per year.

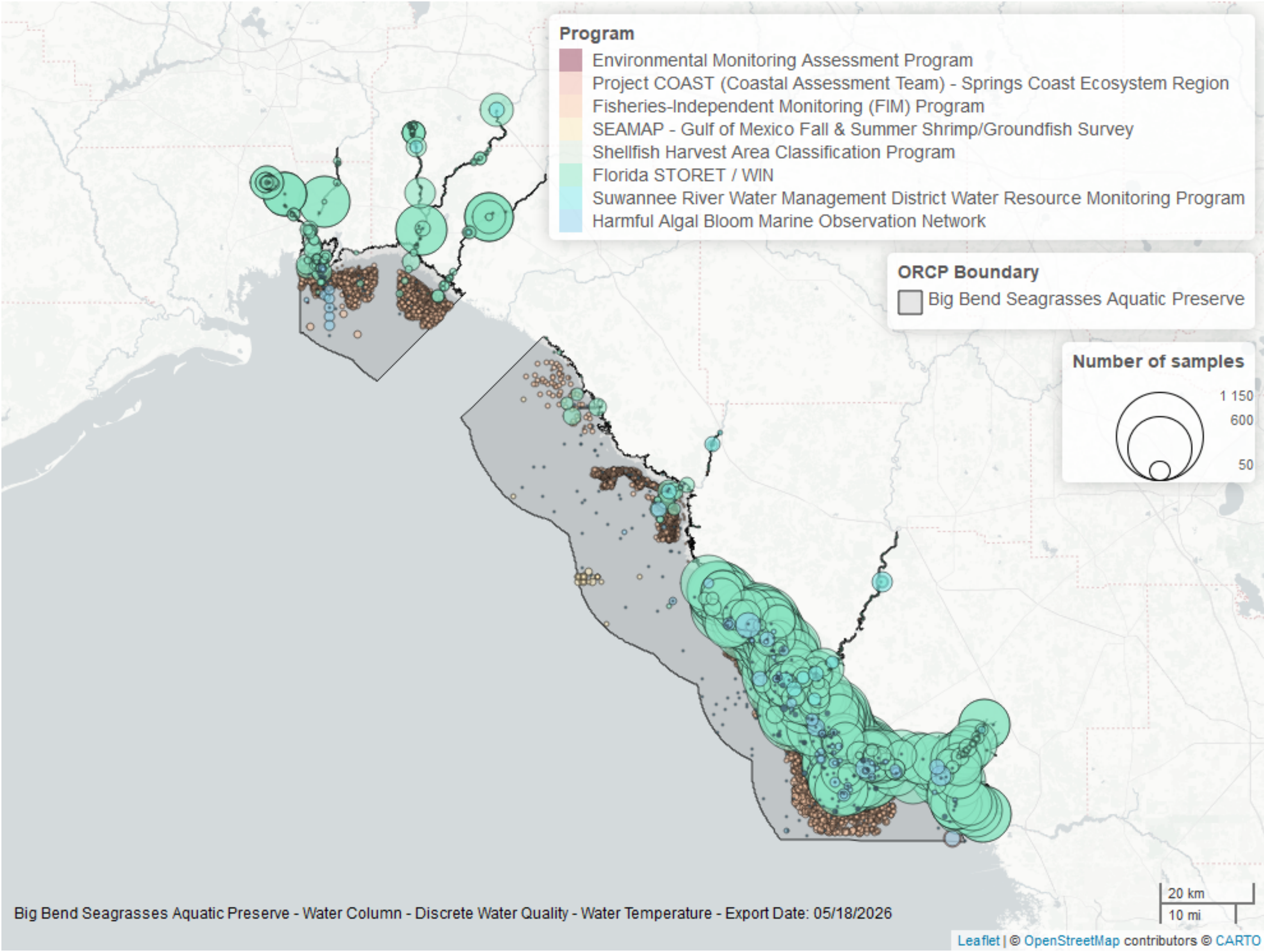


Figure 22: Map showing location of discrete water quality sampling locations within the boundaries of *Big Bend Seagrasses Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Table 27: Programs contributing data for Water Temperature

| ProgramID | N_Data | YearMin | YearMax |
|---|---|---|---|
| 5002 | 100626 | 1989 | 2025 |
| 69 | 57445 | 1996 | 2024 |
| 95 | 1376 | 1964 | 2018 |
| 477 | 398 | 2016 | 2025 |
| 540 | 133 | 2017 | 2022 |
| 60 | 119 | 1986 | 2014 |
| 5008 | 53 | 2021 | 2025 |
| 115 | 43 | 1991 | 2004 |

**Program names:**

*5002* - Florida STORET / WIN[1]
*69* - Fisheries-Independent Monitoring (FIM) Program[9]
*95* - Harmful Algal Bloom Marine Observation Network[10]
*477* - Suwannee River Water Management District Water Resource Monitoring Program[3]

*540* - Shellfish Harvest Area Classification Program[4]
*60* - Southeast Area Monitoring and Assessment Program (SEAMAP) - Gulf of Mexico Fall & Summer Shrimp/Groundfish Survey[11]
*5008* - Project COAST (Coastal Assessment Team) - Springs Coast Ecosystem Region[5]
*115* - Environmental Monitoring Assessment Program[8]

# Water Quality - Continuous

The following files were used in the continuous analysis:

- *Combined_WQ_WC_NUT_cont_Chlorophyll_a_Uncorrected_for_Pheophytin-2026-Mar-06.txt*
- *Combined_WQ_WC_NUT_cont_Dissolved_Oxygen-2026-Mar-06.txt*
- *Combined_WQ_WC_NUT_cont_Dissolved_Oxygen_Saturation-2026-Mar-06.txt*
- *Combined_WQ_WC_NUT_cont_Fluorescent_Dissolved_Organic_Matter-2026-Mar-06.txt*
- *Combined_WQ_WC_NUT_cont_pH-2026-Mar-06.txt*
- *Combined_WQ_WC_NUT_cont_Salinity-2026-Mar-06.txt*
- *Combined_WQ_WC_NUT_cont_Specific_Conductivity-2026-Mar-06.txt*
- *Combined_WQ_WC_NUT_cont_Turbidity-2026-Mar-06.txt*
- *Combined_WQ_WC_NUT_cont_Water_Temperature-2026-Mar-06.txt*

### Continuous monitoring locations in Big Bend Seagrasses Aquatic Preserve

Table 28: Station overview for Continuous parameters by Program

| *ProgramID* | *ProgramLocationID* | *Years of Data* | *Use in Analysis* | *Parameters* |
|---|---|---|---|---|
| 7 | 02313700 | 8 | TRUE | pH , Salinity |
| 7 | 02313700 | 9 | TRUE | Dissolved Oxygen , Specific Conductivity |
| 7 | 02313700 | 25 | TRUE | Water Temperature |
| 7 | 02323566 | 12 | TRUE | pH |
| 7 | 02323566 | 13 | TRUE | Dissolved Oxygen , Water Temperature , Specific Conductivity |
| 7 | 02323592 | 16 | TRUE | Salinity |
| 7 | 02323592 | 27 | TRUE | Water Temperature , Specific Conductivity |
| 7 | 02326050 | 6 | TRUE | Water Temperature , Specific Conductivity |
| 7 | 02326516 | 6 | TRUE | pH |
| 7 | 02326516 | 7 | TRUE | Dissolved Oxygen , Water Temperature , Specific Conductivity |
| 7 | 02326526 | 10 | TRUE | pH , Dissolved Oxygen , Water Temperature , Specific Conductivity |
| 7 | 02326550 | 6 | TRUE | Specific Conductivity |
| 7 | 02326550 | 8 | TRUE | Salinity |
| 7 | 02326550 | 25 | TRUE | Water Temperature |
| 7 | 02327022 | 3 | FALSE | Water Temperature , Specific Conductivity |
| 7 | 291652083064100 | 1 | FALSE | Water Temperature, Salinity |
| 7 | 291842083085100 | 1 | FALSE | Water Temperature, Salinity |
| 471 | BBSCK | 1 | FALSE | pH , Dissolved Oxygen Saturation, Dissolved Oxygen , Turbidity , Water Temperature , Salinity , Specific Conductivity |
| 471 | BBSDB | 10 | TRUE | pH , Dissolved Oxygen Saturation, Dissolved Oxygen , Turbidity , Water Temperature , Salinity , Specific Conductivity |
| 471 | BBSSK | 12 | TRUE | pH , Dissolved Oxygen Saturation, Dissolved Oxygen , Turbidity , Water Temperature , Salinity , Specific Conductivity |
| 471 | BBSST | 6 | TRUE | pH , Dissolved Oxygen Saturation, Dissolved Oxygen , Turbidity , Water Temperature , Salinity , Specific Conductivity |
| 471 | BBSSW | 8 | TRUE | pH , Dissolved Oxygen Saturation, Dissolved Oxygen , Turbidity , Water Temperature , Salinity , Specific Conductivity |

### Program names:

*7* - National Water Information System[13]
*471* - Big Bend Seagrasses Aquatic Preserves Continuous Water Quality Monitoring[14]

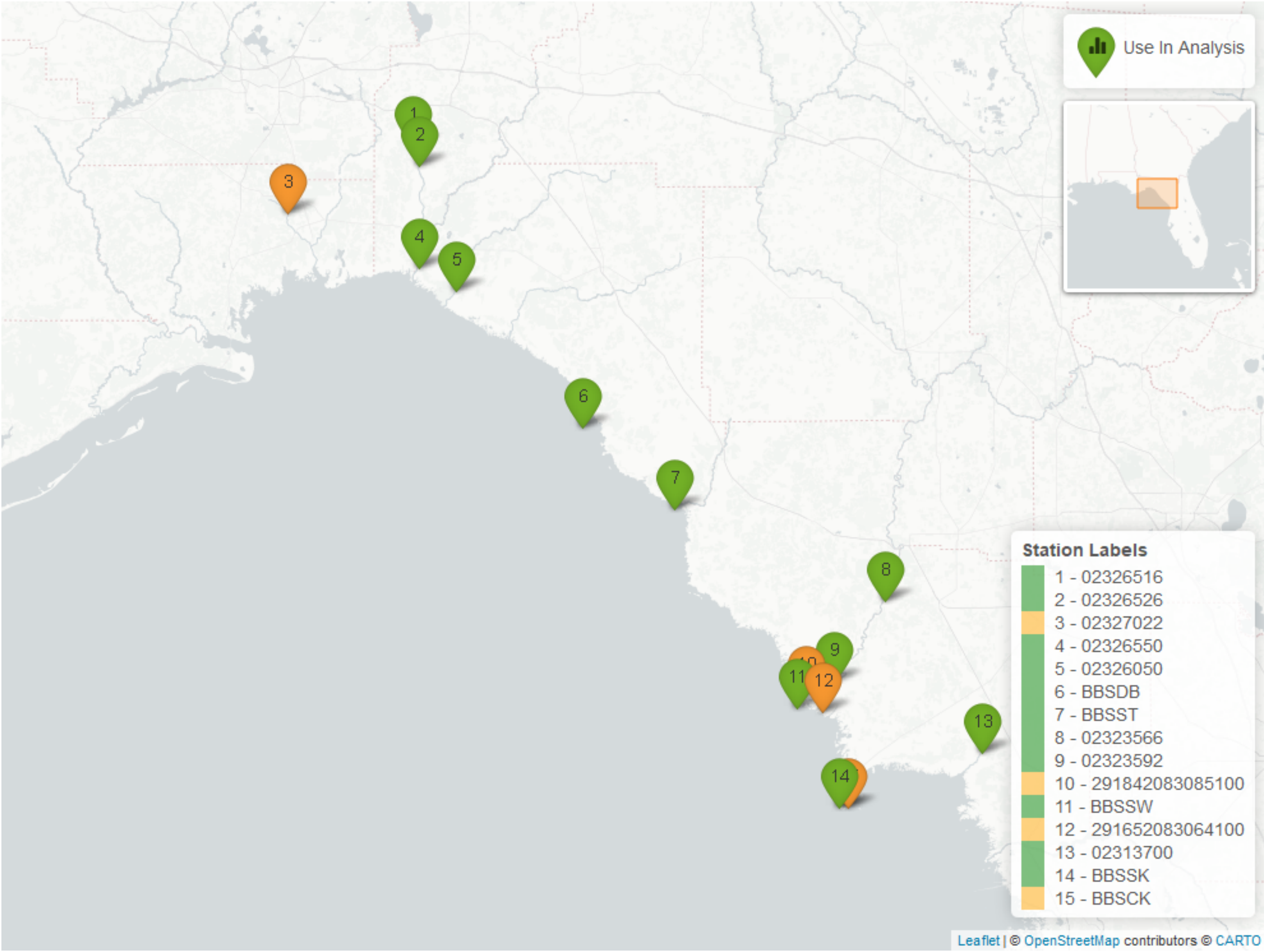


Figure 23: Map showing continuous water quality sampling locations within the boundaries of *Big Bend Seagrasses Aquatic Preserve*. Sites marked as *Use In Analysis* (green) are featured in this report.

**pH - Continuous**

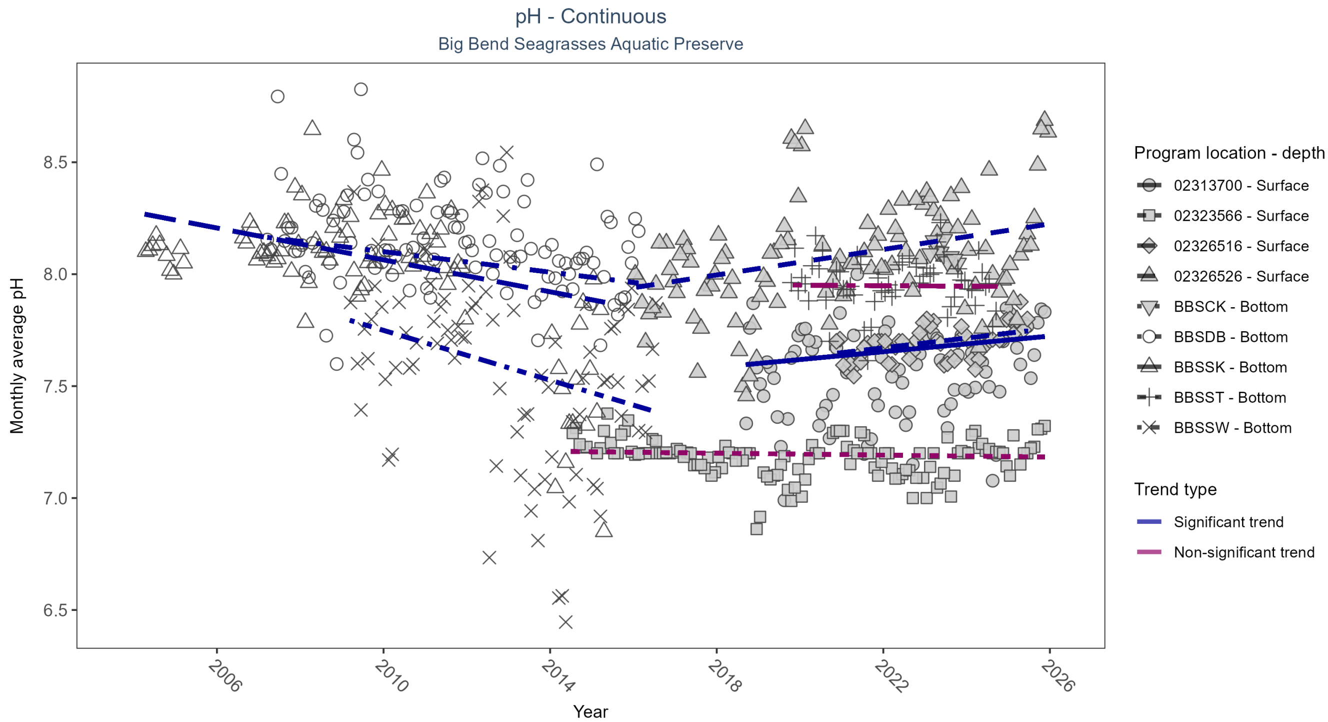


Figure 24: Scatter plot of monthly average pH over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 29: Seasonal Kendall-Tau Results for pH - All Stations

| Program Location | Statistical Trend | No. of Samples | No. Years with Data | Period of Record | Median Result Value | Tau | Sen Intercept | Sen Slope | P |
|---|---|---|---|---|---|---|---|---|---|
| 02313700 | Significantly increasing trend | 2470 | 8 | 2018 - 2025 | 7.6 | 0.22 | 7.58 | 0.02 | 0.02 |
| 02323566 | No significant trend | 3751 | 12 | 2014 - 2025 | 7.2 | -0.08 | 7.21 | 0 | 0.25 |
| 02326516 | Significantly increasing trend | 1485 | 6 | 2020 - 2025 | 7.7 | 0.4 | 7.63 | 0.02 | 0 |
| 02326526 | Significantly increasing trend | 3274 | 10 | 2016 - 2025 | 8.0 | 0.26 | 7.94 | 0.03 | 0 |
| BBSCK | Insufficient data to calculate trend | 18185 | 1 | 2023 - 2023 | 8.1 | - | - | - | - |
| BBSDB | Significantly decreasing trend | 250183 | 10 | 2007 - 2016 | 8.1 | -0.28 | 8.17 | -0.02 | 0 |
| BBSSK | Significantly decreasing trend | 168278 | 10 | 2004 - 2015 | 8.1 | -0.37 | 8.28 | -0.04 | 0 |
| BBSST | No significant trend | 153335 | 6 | 2019 - 2024 | 8.0 | -0.01 | 7.95 | 0 | 0.94 |
| BBSSW | Significantly decreasing trend | 224733 | 8 | 2009 - 2016 | 7.6 | -0.29 | 7.8 | -0.06 | 0 |

At three program locations, monthly average pH increased between 0.02 and 0.03 pH units per year. At three program locations, monthly average pH decreased between 0.02 and 0.06 pH units per year. No detectable change in monthly average pH was observed at two locations. There was insufficient data to fit a model for one location.

### Dissolved Oxygen Saturation - Continuous

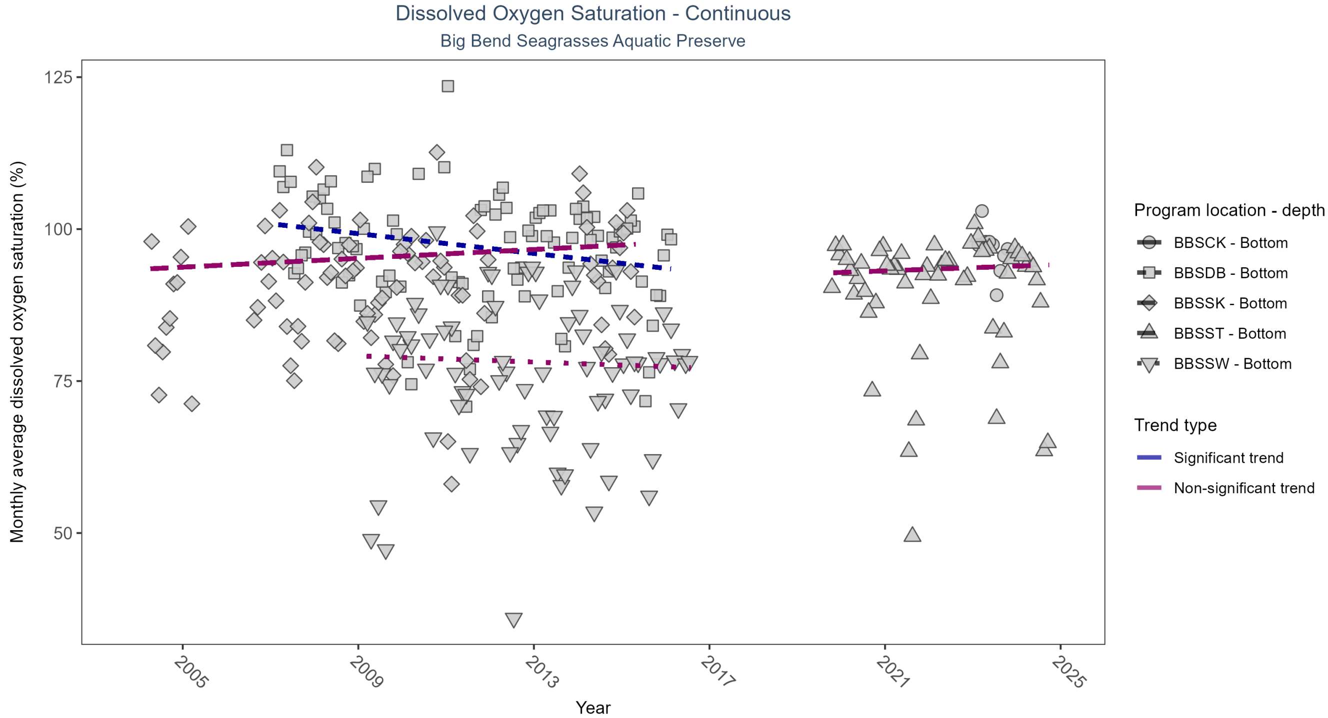


Figure 25: Scatter plot of monthly average dissolved oxygen saturation over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 30: Seasonal Kendall-Tau Results for Dissolved Oxygen Saturation - All Stations

| Program Location | Statistical Trend | No. of Samples | No. Years with Data | Period of Record | Median Result Value | Tau | Sen Intercept | Sen Slope | P |
|---|---|---|---|---|---|---|---|---|---|
| BBSCK | Insufficient data to calculate trend | 19834 | 1 | 2023 - 2023 | 95.7 | - | - | - | - |
| BBSDB | Significantly decreasing trend | 183530 | 10 | 2007 - 2016 | 97.6 | -0.3 | 100.88 | -0.81 | 0 |
| BBSSK | No significant trend | 134196 | 10 | 2004 - 2015 | 91.8 | 0.12 | 93.37 | 0.36 | 0.26 |
| BBSST | No significant trend | 149203 | 6 | 2019 - 2024 | 93.3 | 0.08 | 92.61 | 0.26 | 0.5 |
| BBSSW | No significant trend | 182158 | 8 | 2009 - 2016 | 75.8 | -0.05 | 79.14 | -0.25 | 0.68 |

At one program location, monthly average dissolved oxygen saturation decreased by 0.81% per year. No detectable change in monthly average dissolved oxygen saturation was observed at three locations. There was insufficient data to fit a model for one location.

## Dissolved Oxygen - Continuous

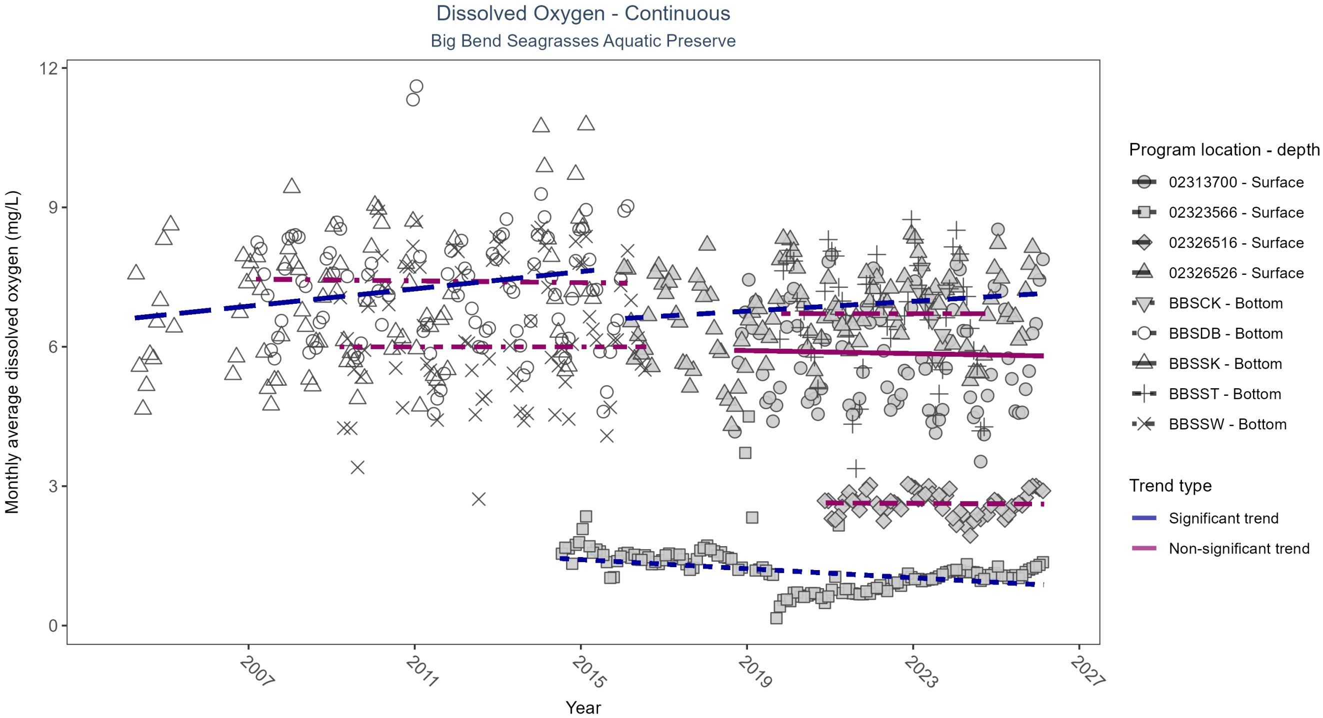


Figure 26: Scatter plot of monthly average dissolved oxygen over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 31: Seasonal Kendall-Tau Results for Dissolved Oxygen - All Stations

| Program Location | Statistical Trend | No. of Samples | No. Years with Data | Period of Record | Median Result Value | Tau | Sen Intercept | Sen Slope | P |
|---|---|---|---|---|---|---|---|---|---|
| 02313700 | No significant trend | 2604 | 9 | 2018 - 2026 | 5.6 | -0.09 | 5.93 | -0.02 | 0.38 |
| 02323566 | Significantly decreasing trend | 4059 | 13 | 2014 - 2026 | 1.2 | -0.33 | 1.47 | -0.05 | 0 |
| 02326516 | No significant trend | 1764 | 7 | 2020 - 2026 | 2.6 | -0.03 | 2.64 | 0 | 0.95 |
| 02326526 | Significantly increasing trend | 3468 | 10 | 2016 - 2025 | 6.7 | 0.18 | 6.61 | 0.05 | 0.01 |
| BBSCK | Insufficient data to calculate trend | 14788 | 1 | 2023 - 2023 | 6.7 | - | - | - | - |
| BBSDB | No significant trend | 184327 | 10 | 2007 - 2016 | 7.3 | -0.05 | 7.46 | -0.01 | 0.61 |
| BBSSK | Significantly increasing trend | 134287 | 10 | 2004 - 2015 | 7.1 | 0.28 | 6.6 | 0.09 | 0 |
| BBSST | No significant trend | 146149 | 6 | 2019 - 2024 | 6.9 | -0.01 | 6.71 | 0 | 1 |
| BBSSW | No significant trend | 182327 | 8 | 2009 - 2016 | 6.2 | 0 | 6 | 0 | 1 |

At two program locations, monthly average dissolved oxygen increased by 0.05 mg/L per year at one site and by 0.09 mg/L per year at the other. At one program location, monthly average dissolved oxygen decreased by 0.05 mg/L per year. No detectable change in monthly average dissolved oxygen was observed at five locations. There was insufficient data to fit a model for one location.

## Turbidity - Continuous

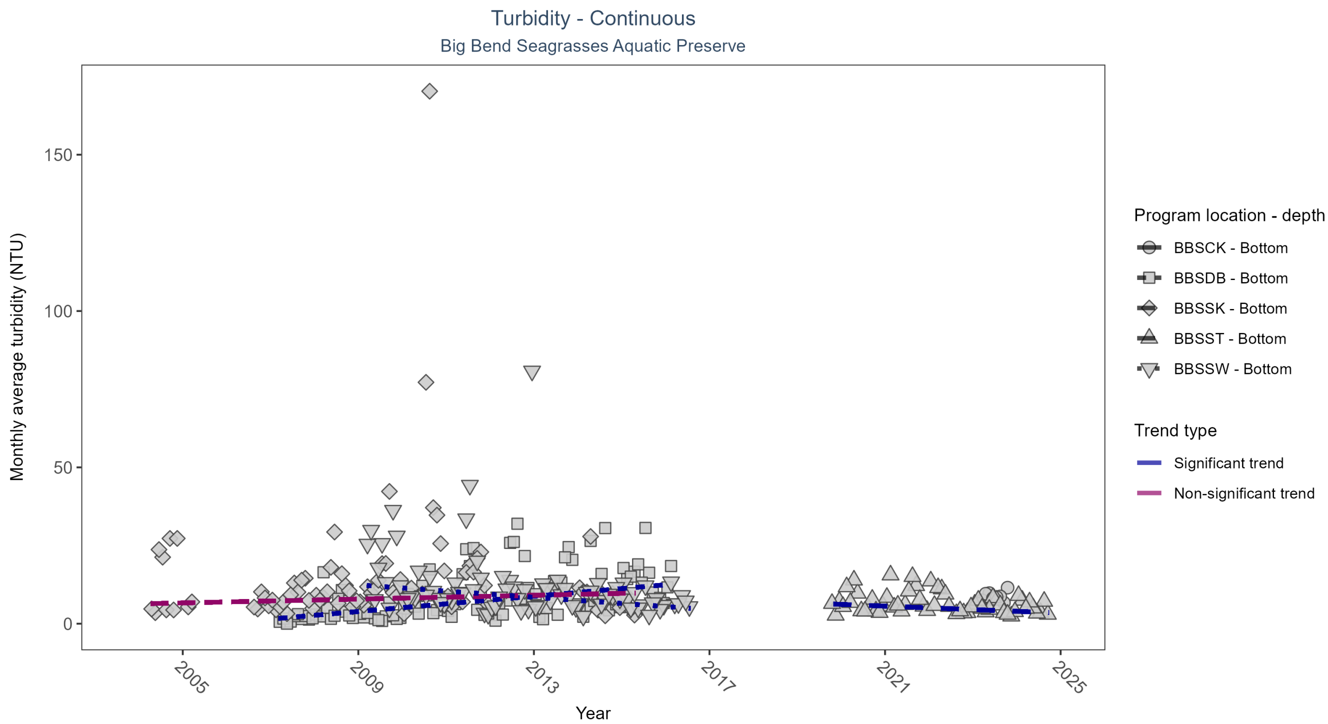


Figure 27: Scatter plot of monthly average turbidity over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 32: Seasonal Kendall-Tau Results for Turbidity - All Stations

| Program Location | Statistical Trend | No. of Samples | No. Years with Data | Period of Record | Median Result Value | Tau | Sen Intercept | Sen Slope | P |
|---|---|---|---|---|---|---|---|---|---|
| BBSCK | Insufficient data to calculate trend | 23243 | 1 | 2023 - 2023 | 7 | - | - | - | - |
| BBSDB | Significantly increasing trend | 224613 | 10 | 2007 - 2016 | 1 | 0.49 | 1.47 | 1.22 | 0 |
| BBSSK | No significant trend | 165043 | 10 | 2004 - 2015 | 5 | 0.11 | 6.35 | 0.3 | 0.19 |
| BBSST | Significantly decreasing trend | 157393 | 6 | 2019 - 2024 | 4 | -0.31 | 6.73 | -0.55 | 0.01 |
| BBSSW | Significantly decreasing trend | 202699 | 8 | 2009 - 2016 | 6 | -0.35 | 12.41 | -0.99 | 0 |

At one program location, monthly average turbidity increased by 1.22 NTU per year. At two program locations, monthly average turbidity decreased by 0.55 NTU per year at one site and by 0.99 NTU per year at the other. No detectable change in monthly average turbidity was observed at one location. There was insufficient data to fit a model for one location.

## Water Temperature - Continuous - Program 7

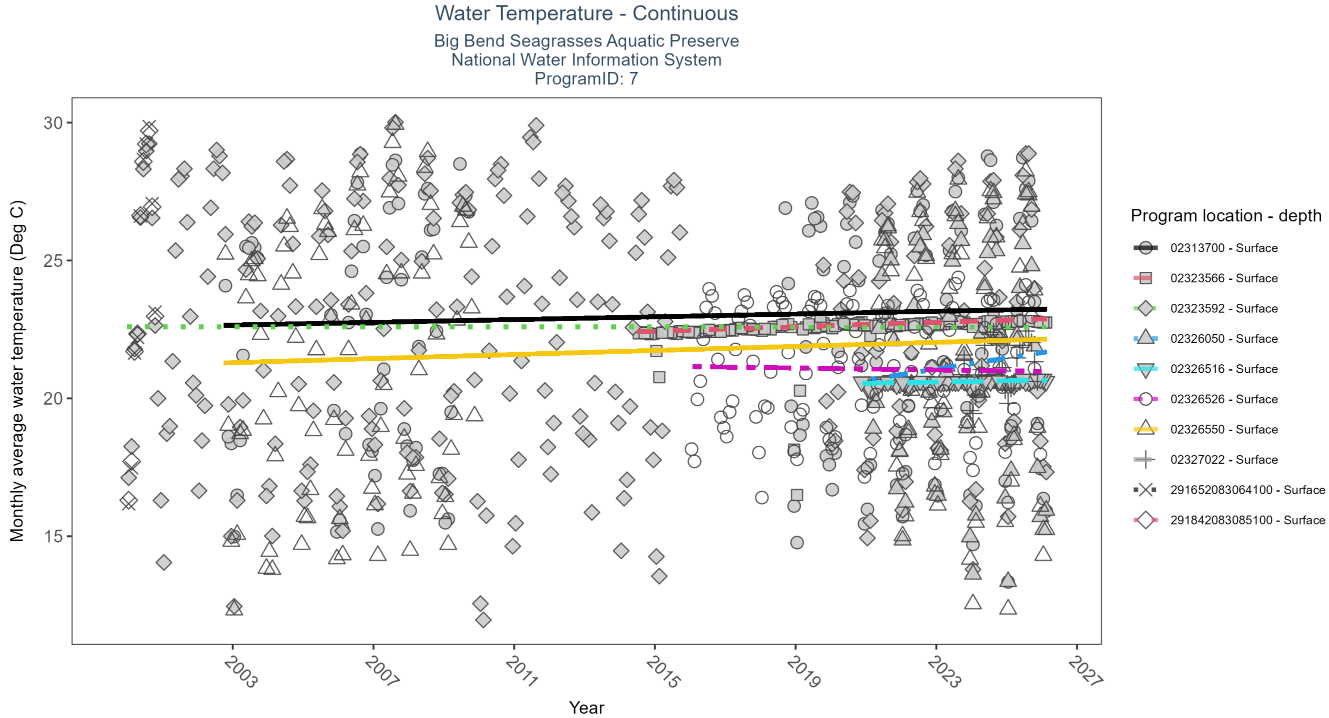


Figure 28: Scatter plot of monthly average water temperature over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 33: Seasonal Kendall-Tau Results for Water Temperature - Program 7

| Program Location | Statistical Trend | No. of Samples | No. Years with Data | Period of Record | Median Result Value | Tau | Sen Intercept | Sen Slope | P |
|---|---|---|---|---|---|---|---|---|---|
| 02313700 | Significantly increasing trend | 4275 | 16 | 2002 - 2026 | 22.9 | 0.14 | 22.63 | 0.03 | 0.03 |
| 02323566 | Significantly increasing trend | 4200 | 13 | 2014 - 2026 | 22.6 | 0.78 | 22.39 | 0.04 | 0 |
| 02323592 | No significant trend | 13260 | 24 | 2000 - 2026 | 23.3 | 0 | 22.6 | 0 | 0.98 |
| 02326050 | No significant trend | 1876 | 6 | 2021 - 2026 | 21.6 | 0.2 | 20.67 | 0.2 | 0.1 |
| 02326516 | Significantly increasing trend | 1797 | 7 | 2020 - 2026 | 20.6 | 0.42 | 20.53 | 0.02 | 0 |
| 02326526 | No significant trend | 3525 | 10 | 2016 - 2025 | 21.5 | -0.08 | 21.15 | -0.02 | 0.29 |
| 02326550 | Significantly increasing trend | 5202 | 14 | 2002 - 2026 | 22.3 | 0.21 | 21.26 | 0.04 | 0 |
| 02327022 | Insufficient data to calculate trend | 743 | 3 | 2023 - 2025 | 21.4 | - | - | - | - |
| 291652083064100 | Insufficient data to calculate trend | 473 | 1 | 2000 - 2000 | 26.1 | - | - | - | - |
| 291842083085100 | Insufficient data to calculate trend | 542 | 1 | 2000 - 2000 | 23.1 | - | - | - | - |

At four program locations, monthly average water temperature increased between 0.02 and 0.04°C per year. At one program location, monthly average water temperature decreased by 0.37°C per year. No detectable change in monthly average water temperature was observed at six locations. There was insufficient data to fit a model for four locations.

### Water Temperature - Continuous - Program 471

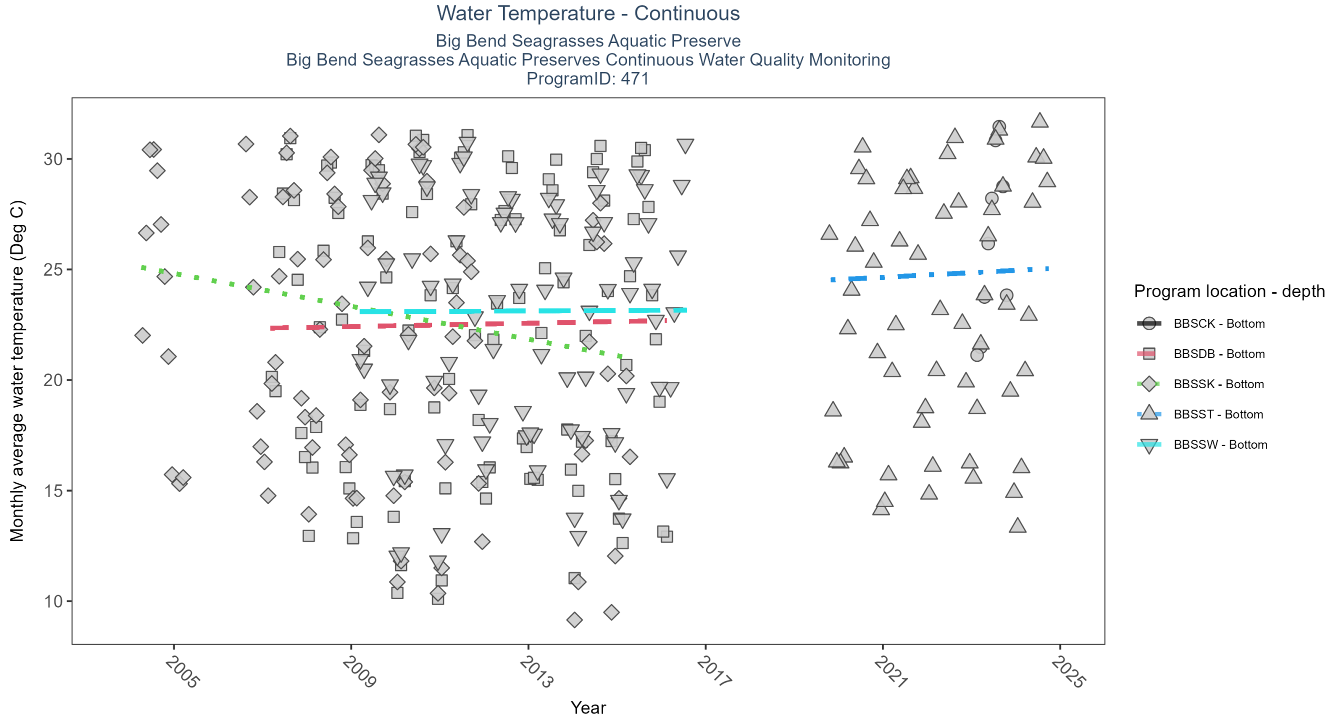


Figure 29: Scatter plot of monthly average water temperature over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 34: Seasonal Kendall-Tau Results for Water Temperature - Program 471

| Program Location | Statistical Trend | No. of Samples | No. Years with Data | Period of Record | Median Result Value | Tau | Sen Intercept | Sen Slope | P |
|---|---|---|---|---|---|---|---|---|---|
| BBSCK | Insufficient data to calculate trend | 22232 | 1 | 2023 - 2023 | 26.4 | - | - | - | - |
| BBSDB | No significant trend | 265988 | 10 | 2007 - 2016 | 23.2 | 0.08 | 22.34 | 0.04 | 0.29 |
| BBSSK | Significantly decreasing trend | 179213 | 10 | 2004 - 2015 | 21.7 | -0.33 | 25.19 | -0.37 | 0 |
| BBSST | No significant trend | 163227 | 6 | 2019 - 2024 | 23.5 | 0.1 | 24.44 | 0.1 | 0.52 |
| BBSSW | No significant trend | 227996 | 8 | 2009 - 2016 | 23.7 | 0.02 | 23.09 | 0.01 | 0.9 |

At four program locations, monthly average water temperature increased between 0.02 and 0.04°C per year. At one program location, monthly average water temperature decreased by 0.37°C per year. No detectable change in monthly average water temperature was observed at six locations. There was insufficient data to fit a model for four locations.

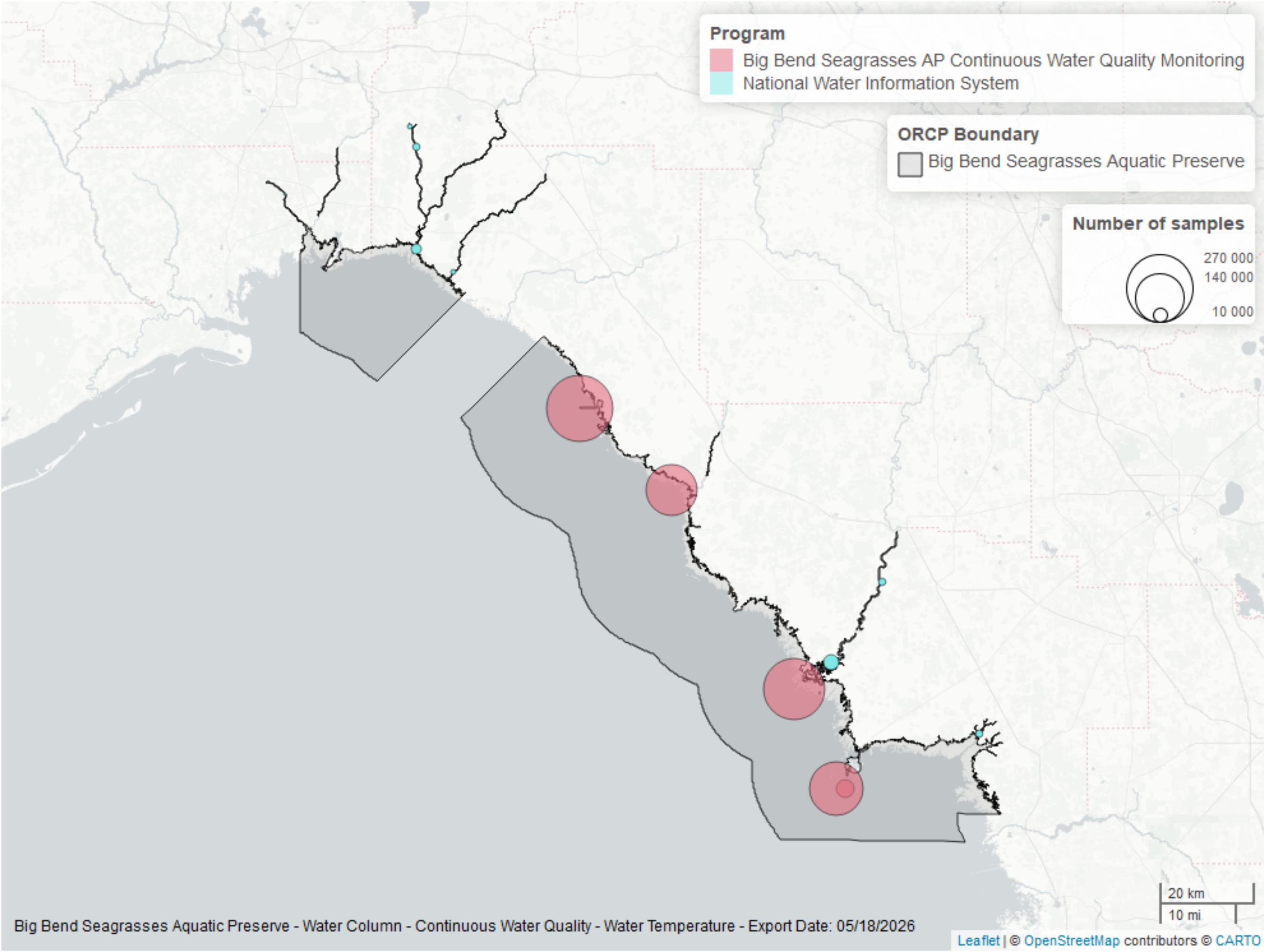


Figure 30: Map showing location of water temperature continuous water quality sampling locations within the boundaries of *Big Bend Seagrasses Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

## Salinity - Continuous

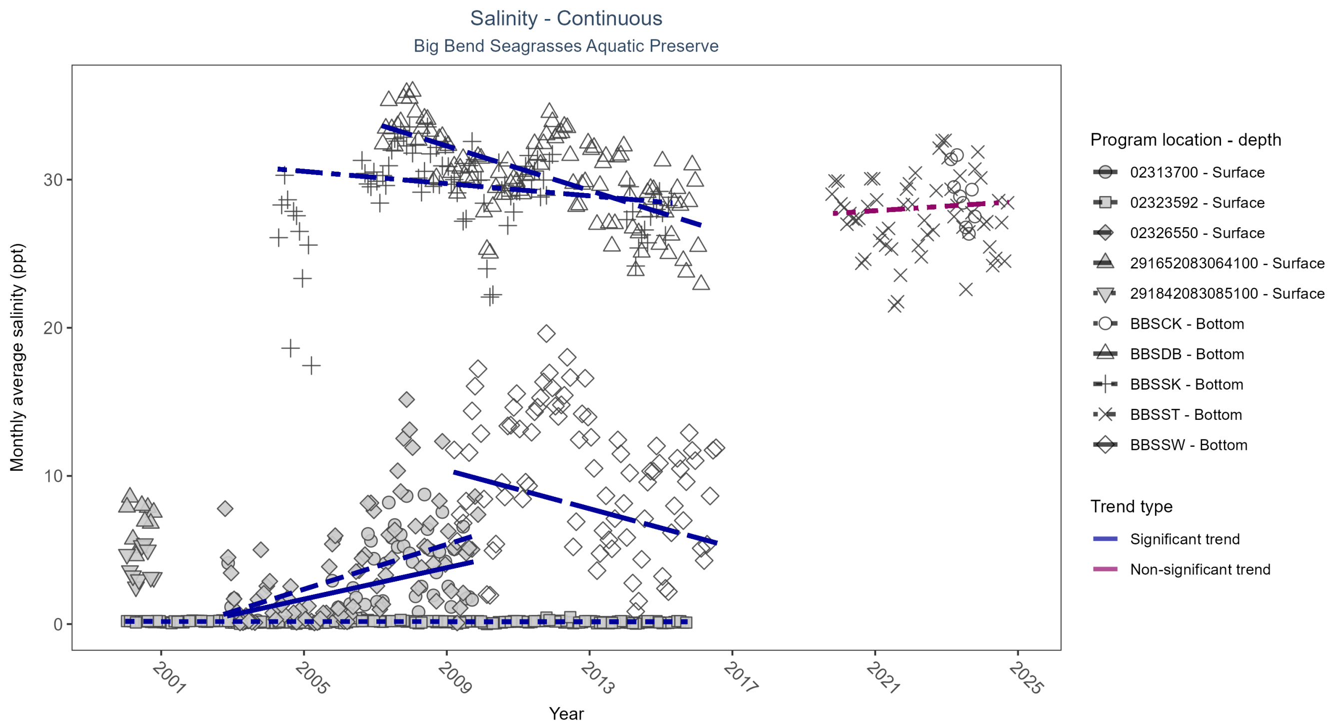


Figure 31: Scatter plot of monthly average salinity over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 35: Seasonal Kendall-Tau Results for Salinity - All Stations

| Program Location | Statistical Trend | No. of Samples | No. Years with Data | Period of Record | Median Result Value | Tau | Sen Intercept | Sen Slope | P |
|---|---|---|---|---|---|---|---|---|---|
| 02313700 | Significantly increasing trend | 1601 | 7 | 2002 - 2009 | 2.1 | 0.52 | 0.1 | 0.53 | 0 |
| 02323592 | Significantly decreasing trend | 6064 | 16 | 2000 - 2015 | 0.1 | -0.13 | 0.18 | 0 | 0.02 |
| 02326550 | Significantly increasing trend | 2507 | 8 | 2002 - 2009 | 1.7 | 0.48 | 0.09 | 0.76 | 0 |
| 291652083064100 | Insufficient data to calculate trend | 827 | 1 | 2000 - 2000 | 6.7 | - | - | - | - |
| 291842083085100 | Insufficient data to calculate trend | 584 | 1 | 2000 - 2000 | 3.7 | - | - | - | - |
| BBSCK | Insufficient data to calculate trend | 14782 | 1 | 2023 - 2023 | 29.2 | - | - | - | - |
| BBSDB | Significantly decreasing trend | 265544 | 10 | 2007 - 2016 | 30.6 | -0.53 | 33.78 | -0.75 | 0 |
| BBSSK | Significantly decreasing trend | 178356 | 10 | 2004 - 2015 | 29.6 | -0.22 | 30.77 | -0.21 | 0.02 |
| BBSST | No significant trend | 156720 | 6 | 2019 - 2024 | 28.5 | 0.08 | 27.59 | 0.15 | 0.52 |
| BBSSW | Significantly decreasing trend | 221696 | 8 | 2009 - 2016 | 7.5 | -0.23 | 10.39 | -0.65 | 0.02 |

At two program locations, monthly average salinity increased by 0.53 ppt per year at one site and by 0.76 ppt per year at the other. At four program locations, monthly average salinity decreased between less than 0.01 and 0.75 ppt per year. No detectable change in monthly average salinity was observed at one location. There was insufficient data to fit a model for three locations.

## Specific Conductivity - Continuous - Program 7

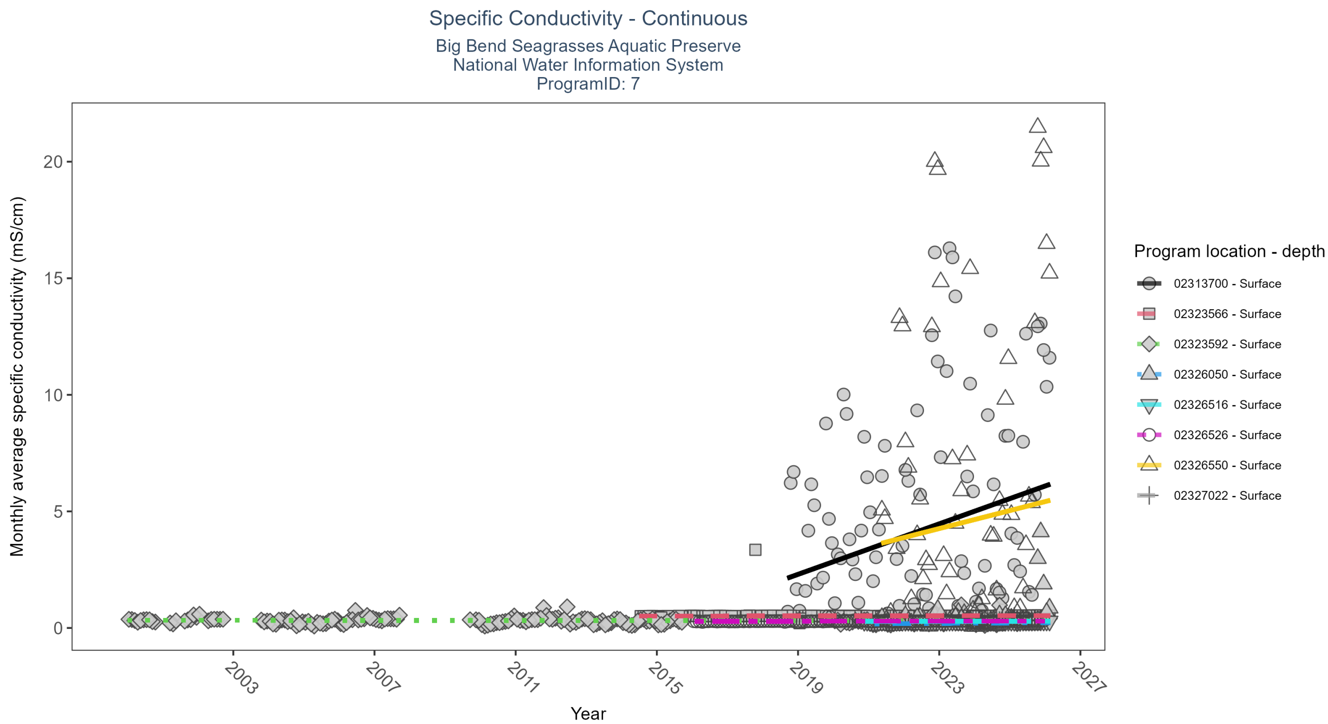


Figure 32: Scatter plot of monthly average specific conductivity over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 36: Seasonal Kendall-Tau Results for Specific Conductivity - Program 7

| Program Location | Statistical Trend | No. of Samples | No. Years with Data | Period of Record | Median Result Value | Tau | Sen Intercept | Sen Slope | P |
|---|---|---|---|---|---|---|---|---|---|
| 02313700 | Significantly increasing trend | 2604 | 9 | 2018 - 2026 | 3.43 | 0.32 | 1.76 | 0.54 | 0 |
| 02323566 | Significantly increasing trend | 4162 | 13 | 2014 - 2026 | 0.52 | 0.46 | 0.51 | 0 | 0 |
| 02323592 | No significant trend | 10826 | 23 | 2000 - 2026 | 0.31 | -0.06 | 0.33 | 0 | 0.22 |
| 02326050 | No significant trend | 2166 | 6 | 2021 - 2026 | 0.24 | 0.12 | 0.17 | 0.01 | 0.36 |
| 02326516 | Significantly increasing trend | 1730 | 7 | 2020 - 2026 | 0.28 | 0.36 | 0.28 | 0 | 0 |
| 02326526 | Significantly increasing trend | 3511 | 10 | 2016 - 2025 | 0.29 | 0.23 | 0.29 | 0 | 0 |
| 02326550 | No significant trend | 3071 | 6 | 2021 - 2026 | 2.49 | 0.14 | 3.49 | 0.39 | 0.27 |
| 02327022 | Insufficient data to calculate trend | 694 | 3 | 2023 - 2025 | 0.32 | - | - | - | - |

At four program locations, monthly average specific conductivity increased between less than 0.01 and 0.54 mS/cm per year. At three program locations, monthly average specific conductivity decreased between 0.28 and 1.06 mS/cm per year. No detectable change in monthly average specific conductivity was observed at four locations. There was insufficient data to fit a model for two locations.

## Specific Conductivity - Continuous - Program 471

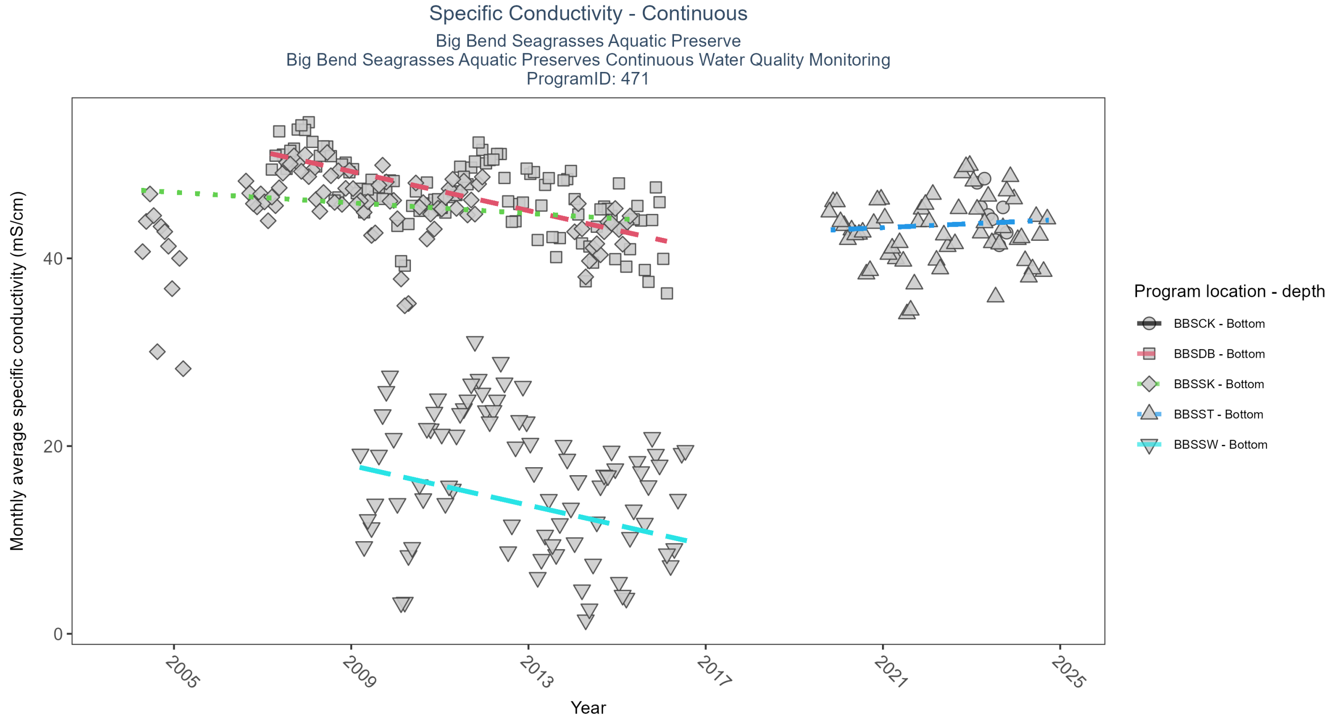


Figure 33: Scatter plot of monthly average specific conductivity over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 37: Seasonal Kendall-Tau Results for Specific Conductivity - Program 471

| Program Location | Statistical Trend | No. of Samples | No. Years with Data | Period of Record | Median Result Value | Tau | Sen Intercept | Sen Slope | P |
|---|---|---|---|---|---|---|---|---|---|
| BBSCK | Insufficient data to calculate trend | 14782 | 1 | 2023 - 2023 | 45.17 | - | - | - | - |
| BBSDB | Significantly decreasing trend | 265988 | 10 | 2007 - 2016 | 47.08 | -0.53 | 51.35 | -1.04 | 0 |
| BBSSK | Significantly decreasing trend | 178359 | 10 | 2004 - 2015 | 45.80 | -0.22 | 47.32 | -0.28 | 0.02 |
| BBSST | No significant trend | 158160 | 6 | 2019 - 2024 | 44.13 | 0.08 | 42.86 | 0.21 | 0.52 |
| BBSSW | Significantly decreasing trend | 221696 | 8 | 2009 - 2016 | 13.01 | -0.23 | 17.94 | -1.06 | 0.02 |

At four program locations, monthly average specific conductivity increased between less than 0.01 and 0.54 mS/cm per year. At three program locations, monthly average specific conductivity decreased between 0.28 and 1.06 mS/cm per year. No detectable change in monthly average specific conductivity was observed at four locations. There was insufficient data to fit a model for two locations.

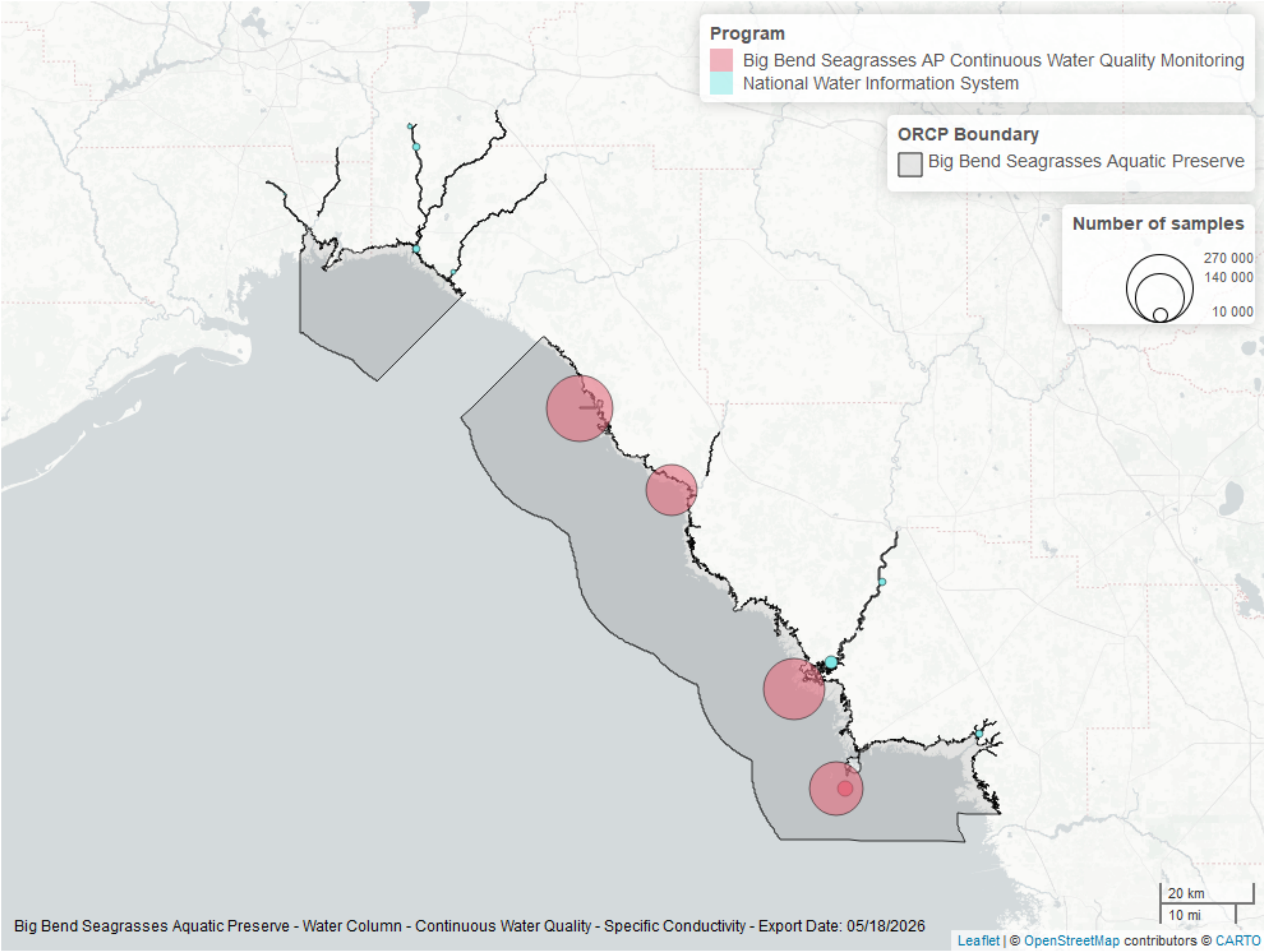


Figure 34: Map showing location of specific conductivity continuous water quality sampling locations within the boundaries of *Big Bend Seagrasses Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

# Submerged Aquatic Vegetation

The data file used is: **All__SAV__Parameters-2026-Jun-04.txt**

**Submerged aquatic vegetation** (*SAV*) refers to plants and plant-like macroalgae species that live entirely underwater. The two primary categories of *SAV* inhabiting Florida estuaries are *benthic macroalgae* and *seagrasses*. They often grow together in dense beds or meadows that carpet the seafloor. *Macroalgae* include multicellular species of green, red and brown algae that often live attached to the substrate by a holdfast. They tend to grow quickly and can tolerate relatively high nutrient levels, making them a threat to seagrasses and other benthic habitats in areas with poor water quality. In contrast, *seagrasses* are grass-like, vascular, flowering plants that are attached to the seafloor by extensive root systems. *Seagrasses* occur throughout the coastal areas of Florida, including protected bays and lagoons as well as deeper offshore waters on the continental shelf. *Seagrasses* have taken advantage of the broad, shallow shelf and clear water to produce two of the most extensive seagrass beds anywhere in continental North America.

## Parameters

**Percent Cover** measures the fraction of an area of seafloor that is covered by SAV, usually estimated by evaluating multiple small areas of seafloor. Percent cover is often estimated for total SAV, individual types of vegetation (seagrass, attached algae, drift algae) and individual species.

**Frequency of Occurrence** was calculated as the number of times a taxon was observed in a year divided by the number of sampling events, multiplied by 100. Analysis is conducted at the quadrat level and is inclusive of all quadrats (i.e., quadrats evaluated using Braun-Blanquet, modified Braun-Blanquet, and percent cover."

## Species

**Turtle grass** (*Thalassia testudinum*) is the largest of the Florida seagrasses, with longer, thicker blades and deeper root structures than any of the other seagrasses. It is considered a climax seagrass species.

**Shoal grass** (*Halodule wrightii*) is an early colonizer of vegetated areas and usually grows in water too shallow for other species except *widgeon grass*. It can often tolerate larger salinity ranges than other seagrass species. *Shoal grass* is characterized by thin, flat blades, that are narrower than *turtle grass* blades.

**Manatee grass** (*Syringodium filiforme*) is easily recognizable because its leaves are thin and cylindrical instead of the flat, ribbon-like form shared by many other seagrass species. The leaves can grow up to half a meter in length. *Manatee grass* is usually found in mixed seagrass beds or small, dense monospecific patches.

**Widgeon grass** (*Ruppia maritima*) grows in both fresh and salt water and is widely distributed throughout Florida's estuaries in less saline areas, particularly in inlets along the east coast. This species resembles *shoal grass* in certain environments but can be identified by the pointed tips of its leaves.

Three species of *Halophila spp.* are found in Florida - **Star grass** (*Halophila engelmannii*), **Paddle grass** (*Halophila decipiens*), and **Johnson's seagrass** (*Halophila johnsonii*). These are smaller, more fragile seagrasses than other Florida species and are considered ephemeral. They grow along a single long rhizome, with short blades. These species are not well-studied, although surveys are underway to define their ecological roles.

### Notes

*Star grass*, *Paddle grass*, and *Johnson's seagrass* will be grouped together and listed as **Halophila spp.** in the following managed areas. This is because several surveys did not specify to the species level:

- Banana River Aquatic Preserve
- Indian River-Malabar to Vero Beach Aquatic Preserve
- Indian River-Vero Beach to Ft. Pierce Aquatic Preserve
- Jensen Beach to Jupiter Inlet Aquatic Preserve
- Loxahatchee River-Lake Worth Creek Aquatic Preserve
- Mosquito Lagoon Aquatic Preserve

- Biscayne Bay Aquatic Preserve
- Florida Keys National Marine Sanctuary

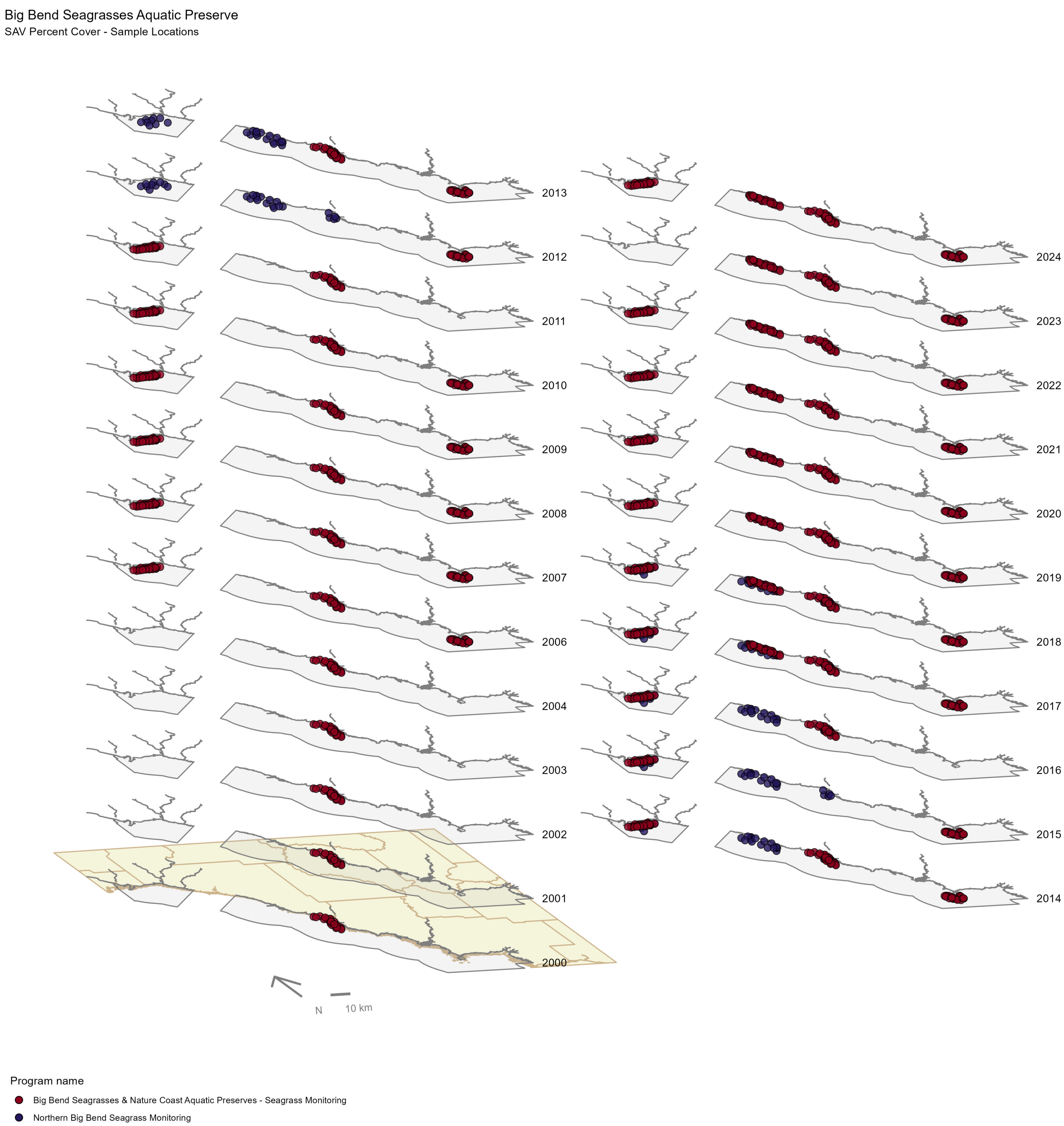


Figure 35: Maps showing the temporal scope of SAV sampling sites within the boundaries of *Big Bend Seagrasses Aquatic Preserve* by program name.

Click here to view spatio-temporal plots on GitHub.

**Sampling locations by Program:**

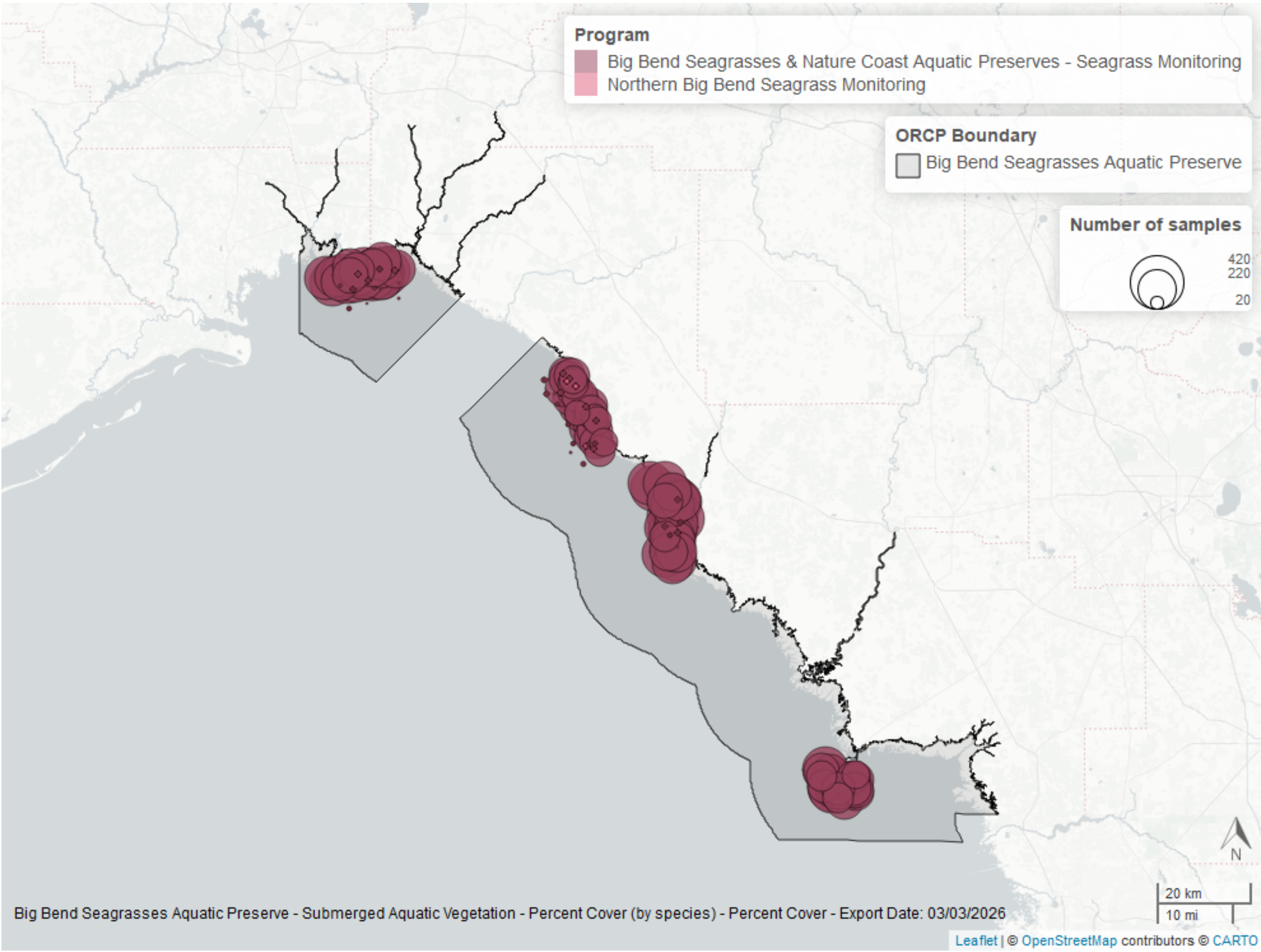


Figure 36: Map showing SAV sampling sites within the boundaries of *Big Bend Seagrasses Aquatic Preserve*. The point size reflects the number of samples at a given sampling site.

Table 38: Program Information for Submerged Aquatic Vegetation

| *ProgramID* | *N-Data* | *YearMin* | *YearMax* | *method* | *Sample Locations* |
|---|---|---|---|---|---|
| 559 | 537 | 2012 | 2018 | Modified Braun Blanquet | 195 |
| 560 | 19753 | 2000 | 2024 | Modified Braun Blanquet | 100 |
| 560 | 4271 | 2022 | 2024 | Percent Cover | 100 |

**Program names:**

*559* - Northern Big Bend Seagrass Monitoring[15]
*560* - Big Bend Seagrasses & Nature Coast Aquatic Preserves - Seagrass Monitoring[12]

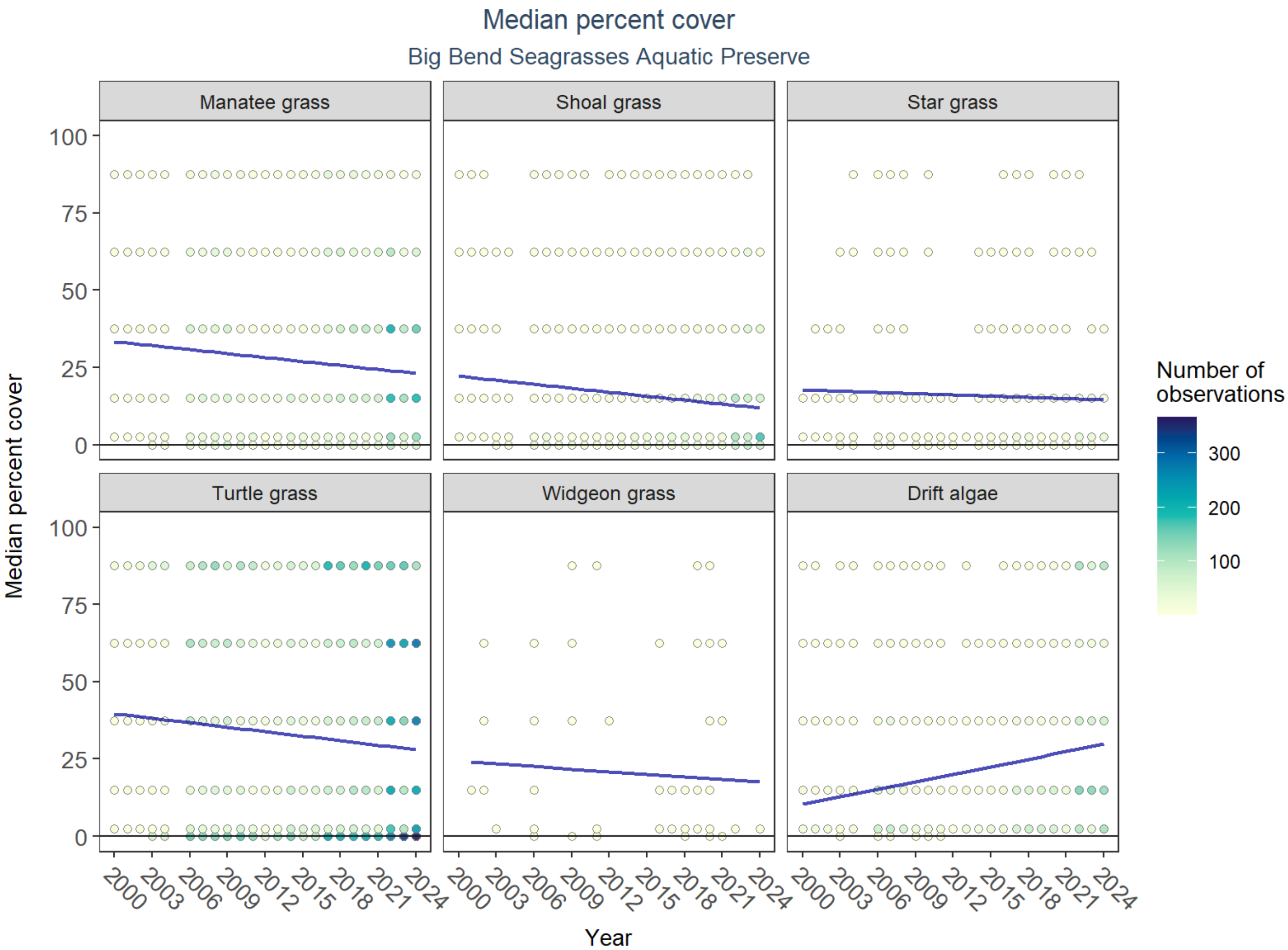


Figure 37: Scatter plots of median percent cover of submerged aquatic vegetation over time by group. Plots for time series that included five or more years of observations show the estimated trend as a blue line.

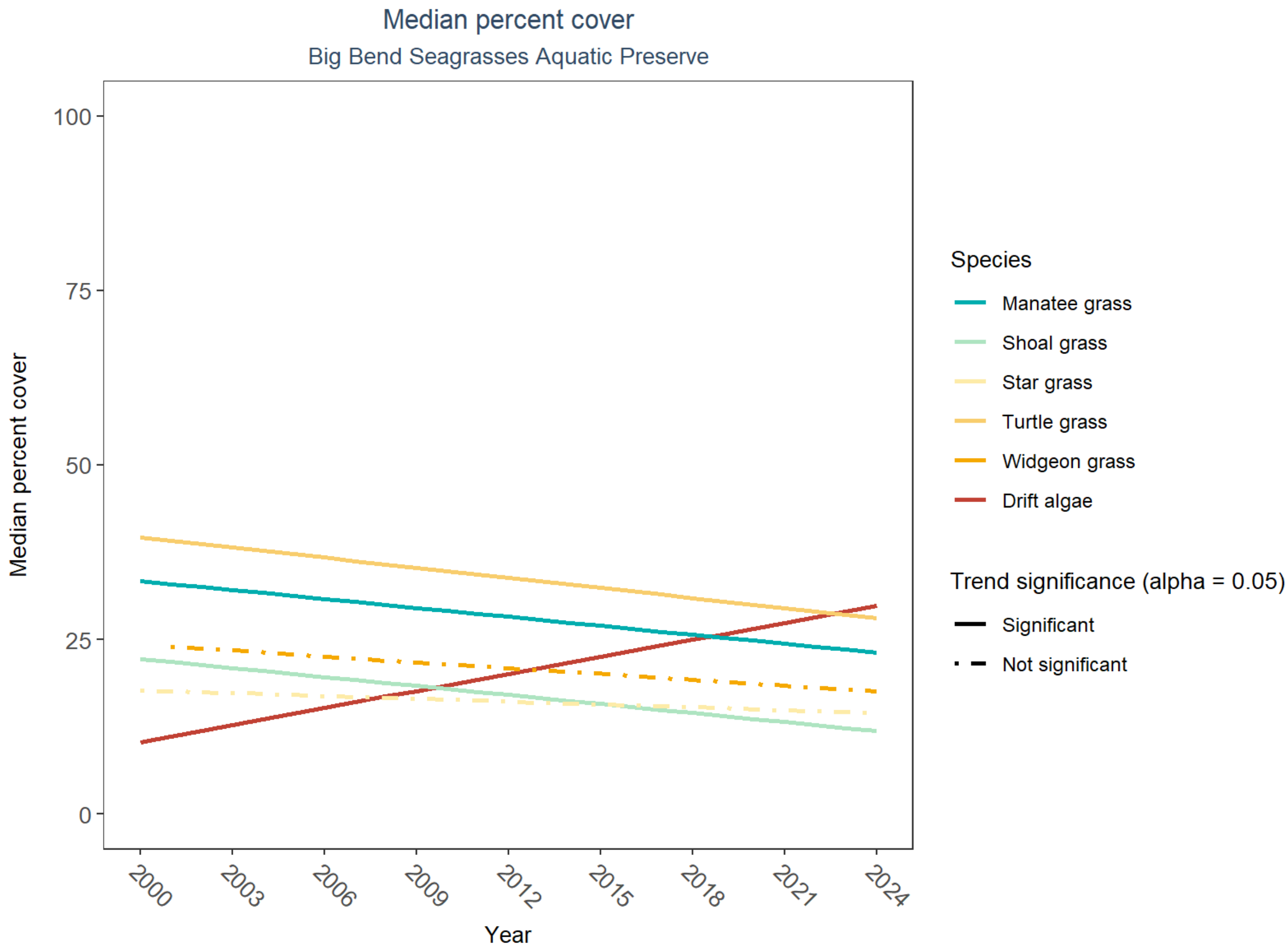


Figure 38: Trend lines of median percent cover of submerged aquatic vegetation over time for species that had five or more years of observations. Line type represents significance (solid) or non-significance (dashed) of the trends.

Table 39: Percent Cover Trend Analysis for Big Bend Seagrasses Aquatic Preserve

| *Species* | *Statistical Trend* | *Period of Record* | *LME Intercept* | *LME Slope* | *P* |
|---|---|---|---|---|---|
| Drift algae | Increasing trend | 2000 - 2024 | 5.453912 | 0.8127329 | 0.0000009 |
| Shoal grass | Decreasing trend | 2000 - 2024 | 24.829746 | -0.4314551 | 0.0001280 |
| Star grass | No detectable trend | 2000 - 2024 | 18.552789 | -0.1347727 | 0.3089913 |
| Widgeon grass | No detectable trend | 2001 - 2024 | 25.939673 | -0.2793284 | 0.5982339 |
| Manatee grass | Decreasing trend | 2000 - 2024 | 35.945396 | -0.4271771 | 0.0000536 |
| Turtle grass | Decreasing trend | 2000 - 2024 | 42.515435 | -0.4814723 | 0.0000000 |

An annual increase in percent cover was observed for drift algae (0.81%). Annual decreases in percent cover were observed for manatee grass (-0.43%), shoal grass (-0.43%), and turtle grass (-0.48%). No detectable change in percent cover was observed for star grass and widgeon grass.

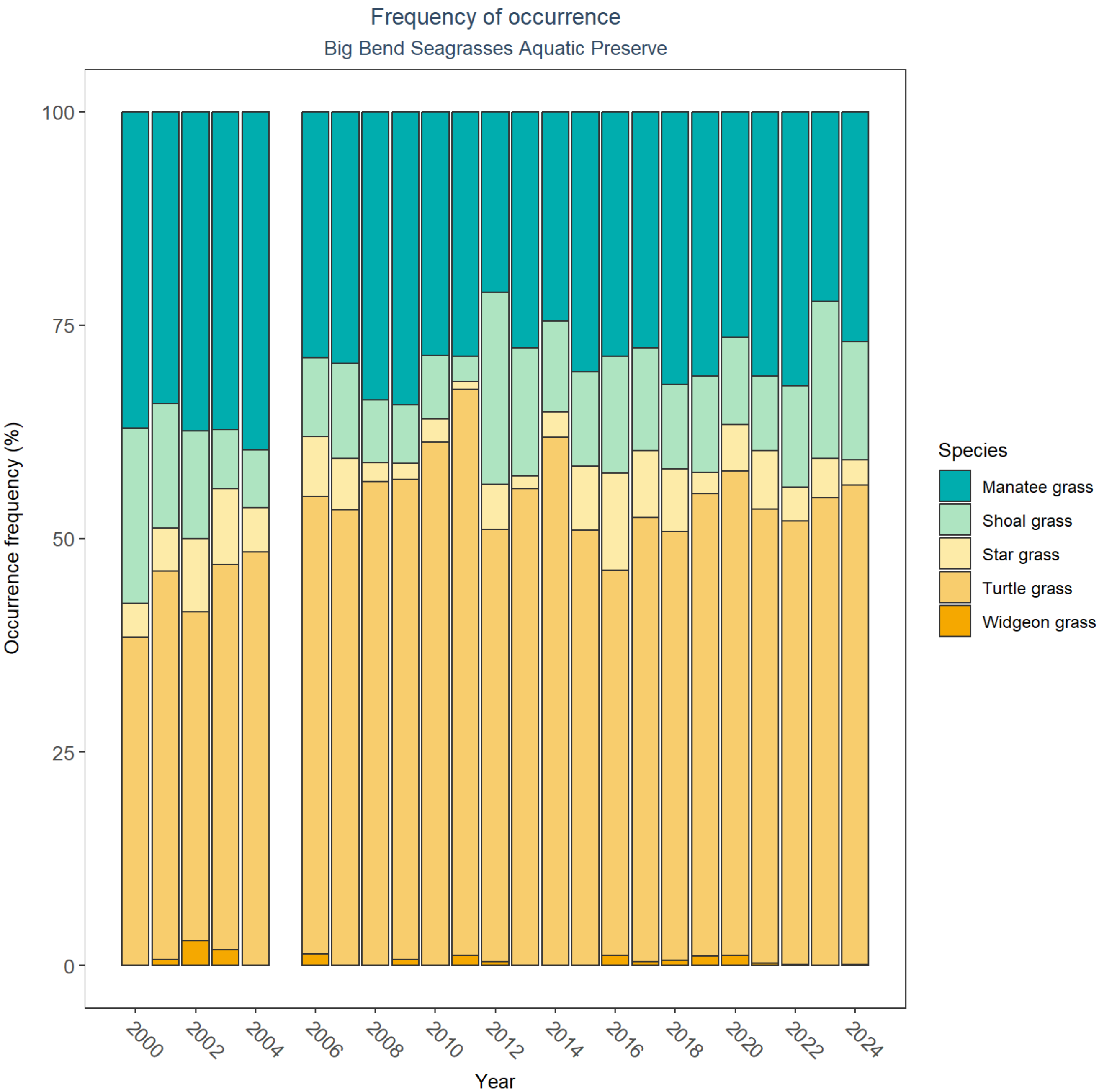


Figure 39: Bar plot of submerged aquatic vegetation species occurrence frequency over time by group.

## SAV Water Column Analysis

The following parameters are available for Big Bend Seagrasses Aquatic Preserve within the SAV_WC_Report:

- Colored Disolved Organic Matter
- Chlorophyll a
- Dissolved Oxygen
- Dissolved Oxygen Saturation
- pH
- Salinity

- Secchi Depth
- Water Temperature
- Total Nitrogen
- Total Suspended Solids
- Turbidity

Access the reports here: DRAFT_SAV_WC_Report.pdf

## Coastal Wetlands

The data file used is: **All__CW__Parameters-2026-Jun-04.txt**

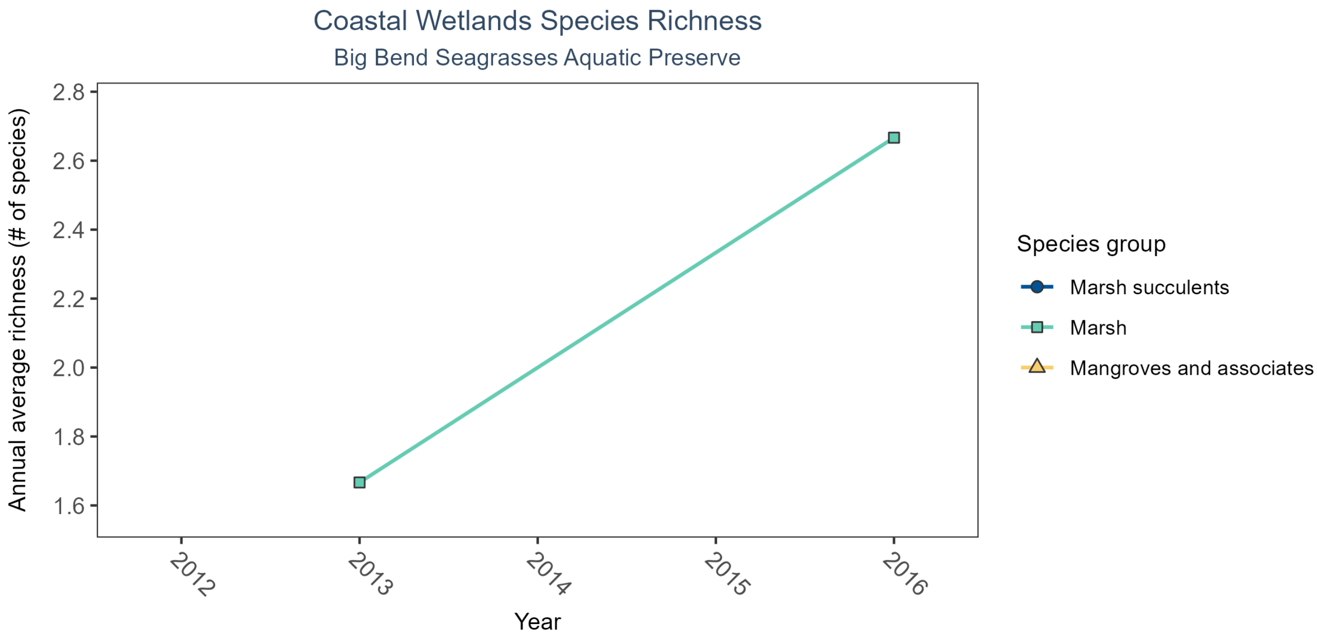


Figure 40: Line graph of annual average coastal wetlands species richness over time for mangroves and associates (triangles), marsh (squares), and marsh succulents (circles). If the time series by species group included more than one year of observations, a line connects data points for visualization.

Table 40: Coastal Wetlands Species Richness

| *Species Group* | *No. of Samples* | *No. Years with Data* | *Period of Record* | *Median No. of Taxa* | *Mean No. of Taxa* |
|---|---|---|---|---|---|
| Marsh | 6 | 2 | 2013 - 2016 | 2 | 2.17 |

Between 2013 and 2016, the median annual number of species for marsh was 2 based on 6 observations.

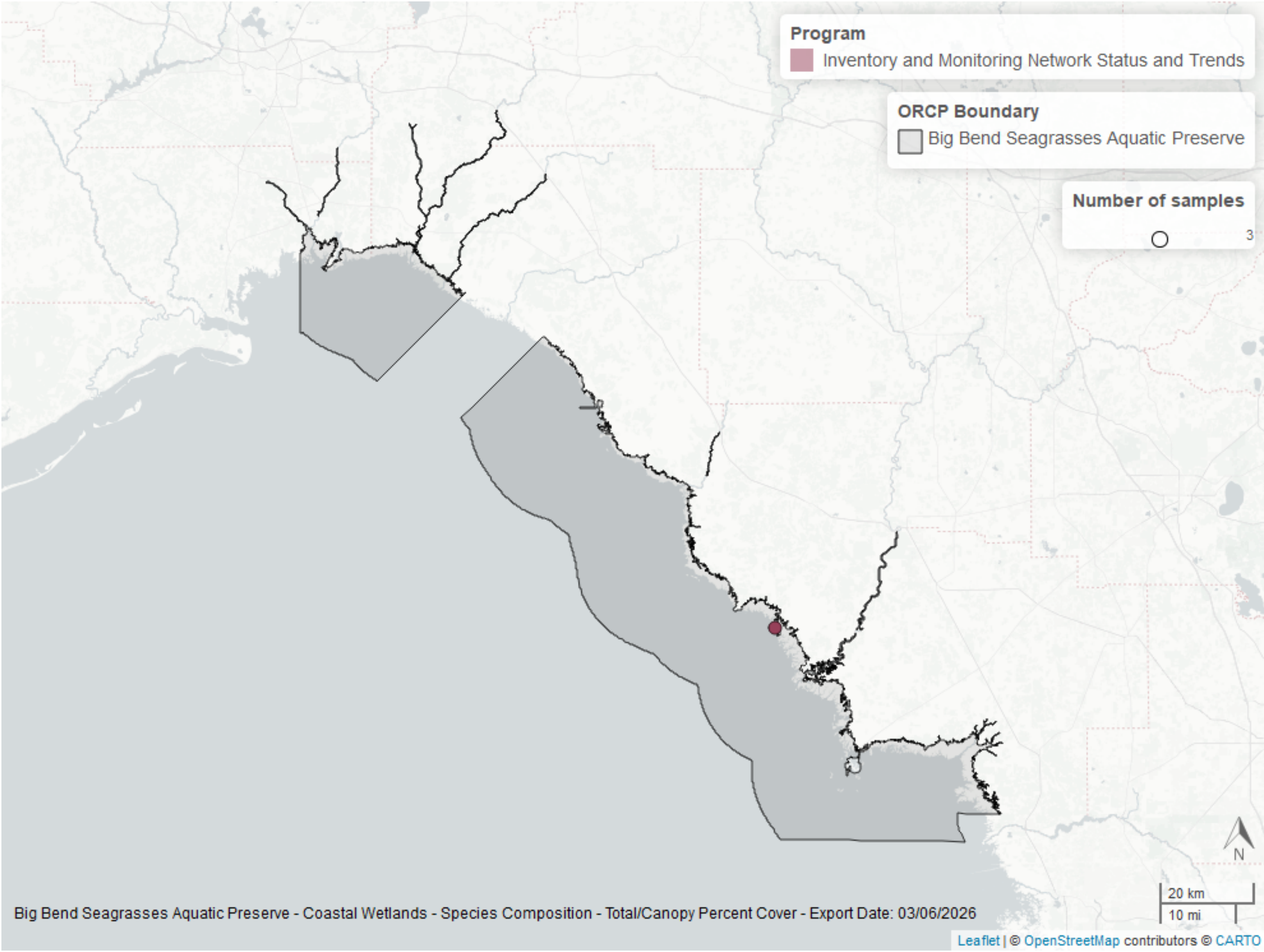


Figure 41: Map showing location of coastal wetlands sampling locations within the boundaries of *Big Bend Seagrasses Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

## Oyster

The data file used is: **All_OYSTER_Parameters-2026-Jun-04.txt**

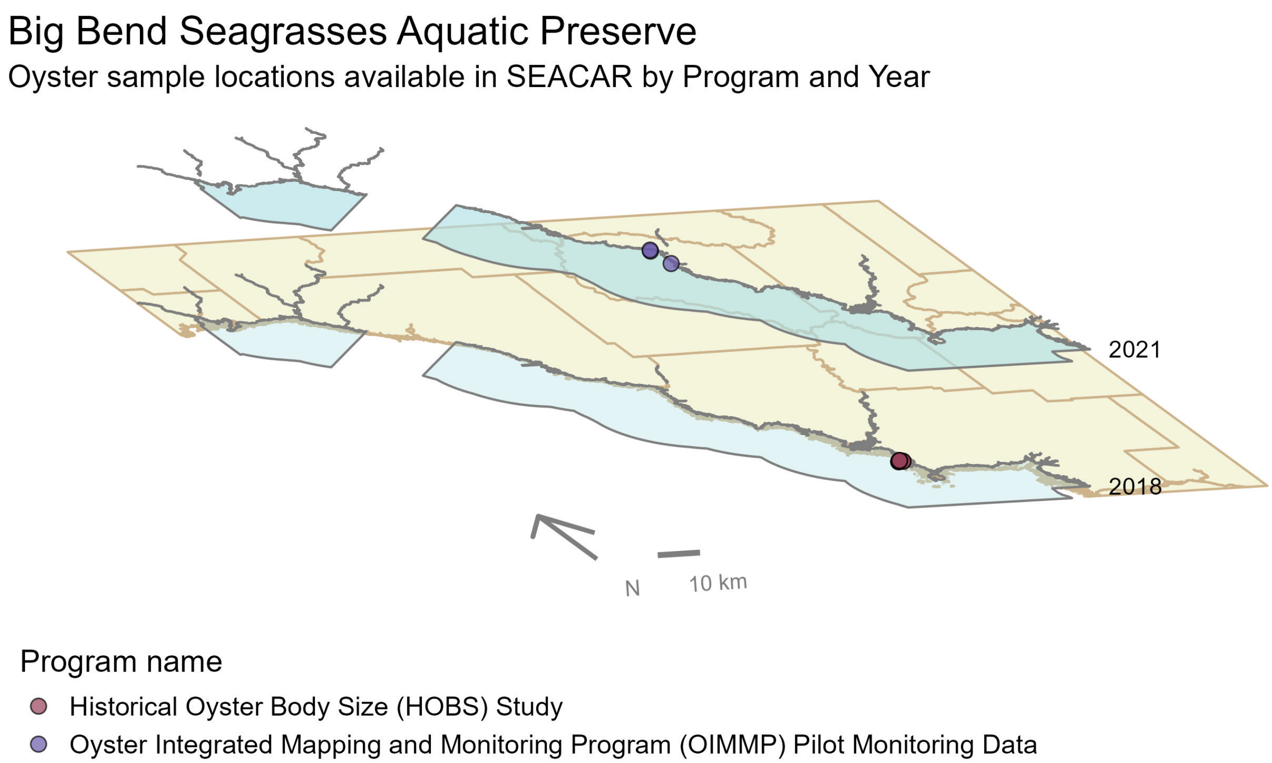


Figure 42: Maps showing the temporal scope of oyster sampling sites within the boundaries of *Big Bend Seagrasses Aquatic Preserve* by program name.

Click here to view spatio-temporal plots on GitHub.

## Shell Height

For natural reefs, there was insufficient data to calculate a trend for live oysters in either the 25-75mm or the >=75mm size class. Models are not run on dead oyster shell measurements.

### Natural

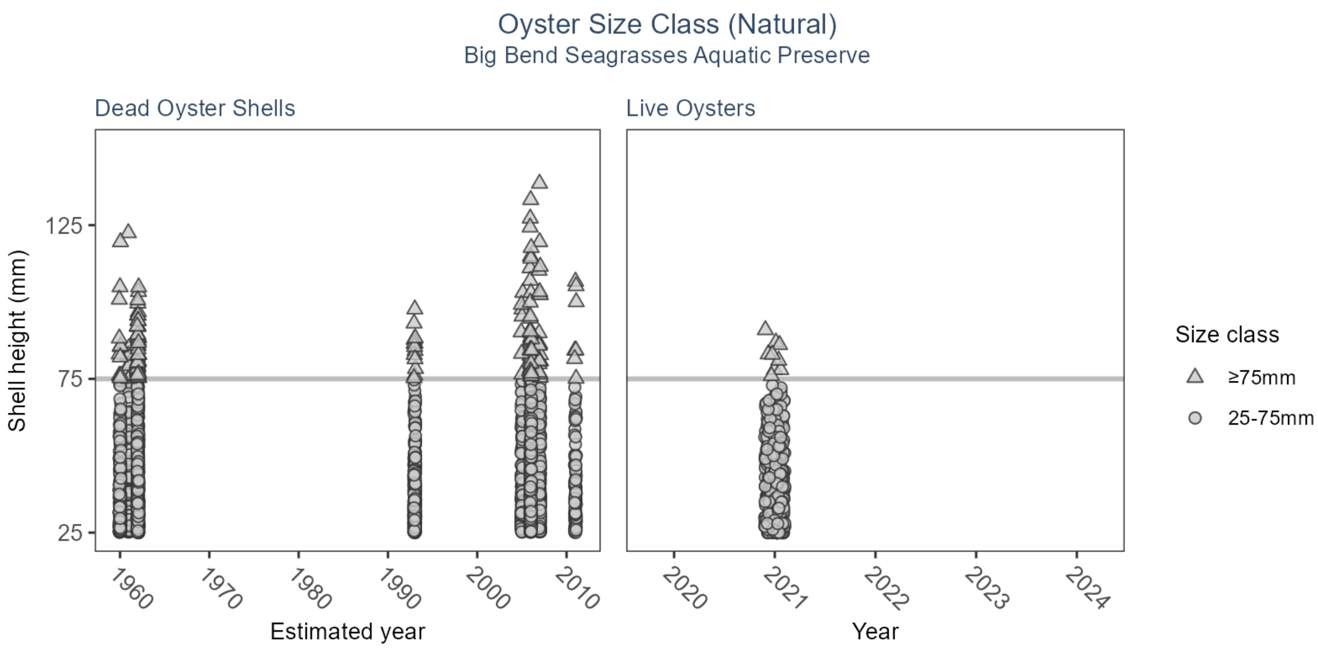


Table 41: Model results for Oyster Shell Height - Natural

| *Habitat Type* | *Shell Type* | *SizeClass* | *Statistical Trend* | *Period of Record* | *No. of Samples* | *Estimate* | *Standard Error* | *Credible Interval* |
|---|---|---|---|---|---|---|---|---|
| Natural | Dead Oyster Shells | >75mm | Model not run on dead oyster shell | 1959 - 2011 | 159 | - | - | - |
| Natural | Dead Oyster Shells | 25-75mm | Model not run on dead oyster shell | 1959 - 2011 | 2012 | - | - | - |
| Natural | Live Oysters | >75mm | Insufficient data | 2021 - 2021 | 9 | - | - | - |
| Natural | Live Oysters | 25-75mm | Insufficient data | 2021 - 2021 | 462 | - | - | - |

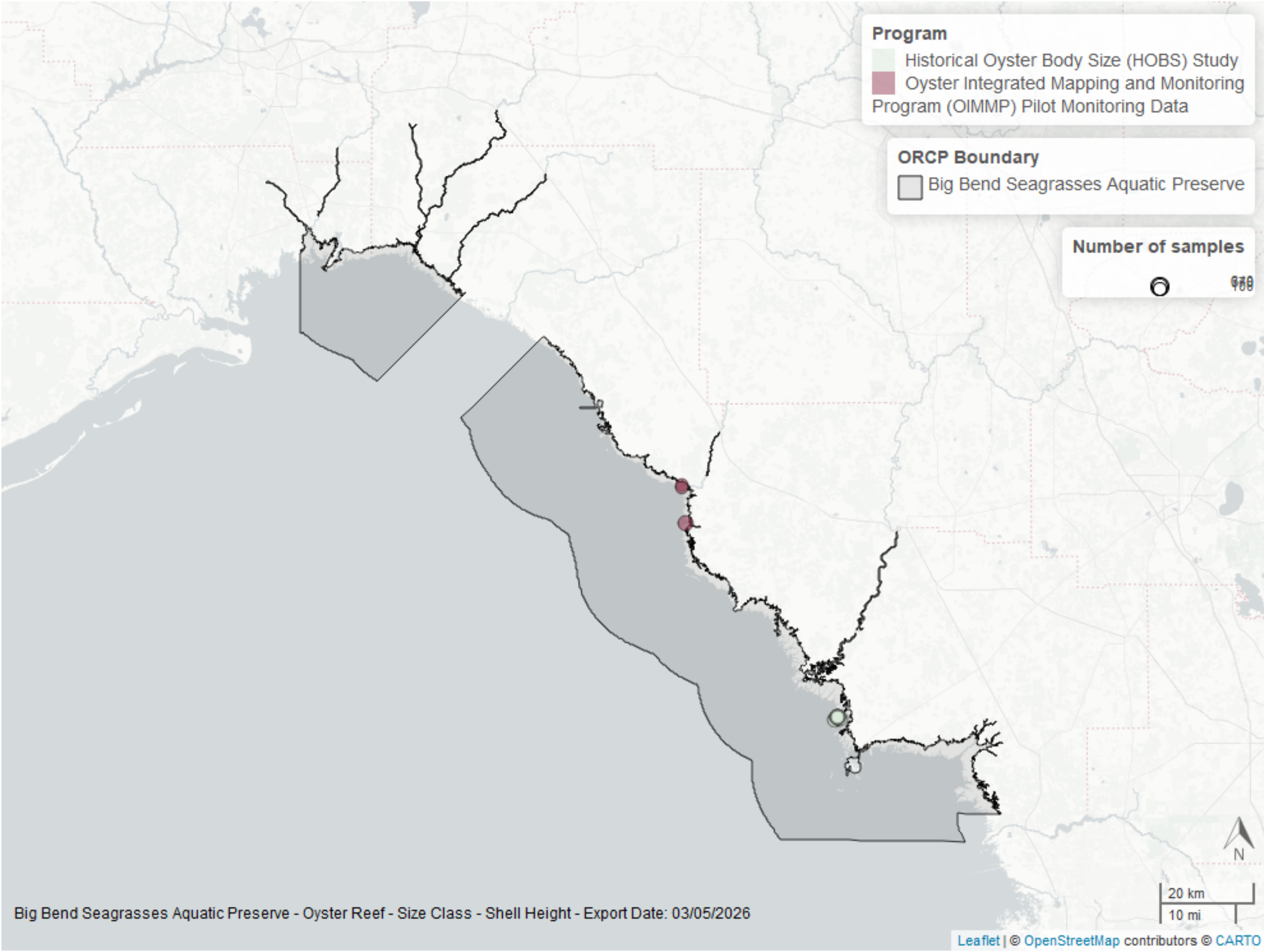


Figure 43: Map showing location of oyster shell height sampling locations within the boundaries of *Big Bend Seagrasses Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

## Species list

| | | |
|---|---|---|
| Acanthophora muscoides[1] | Drift algae[1] | Persicaria hydropiperoides |
| Acetabularia crenulata[1] | Drift red algae[1] | Pluchea camphorata |
| Anadyomene stellata[1] | Eupatorium mikanioides | Polysiphonia sp.[1] |
| Avrainvillea levis[1] | Gracilaria sp.[1] | Ruppia maritima[1] |
| Caulerpa[1] | Halimeda incrassata[1] | Sagittaria graminea |
| Caulerpa ashmeadii[1] | Halodule wrightii[1] | Sagittaria lancifolia lancifolia |
| Caulerpa cupressoides[1] | Halophila engelmannii[1] | Sargassum sp.[1] |
| Caulerpa mexicana[1] | Juncus roemerianus[2] | Sargassum spp.[1] |
| Caulerpa paspaloides[1] | Laurencia spp.[1] | Schoenoplectus tabernaemontani |
| Caulerpa prolifera[1] | Ludwigia repens | Spartina alterniflora[2] |
| Caulerpa racemosa[1] | Lythrum lineare | Spyridia[1] |
| Caulerpa sertularioides[1] | No grass in quadrat[1] | Syringodium filiforme[1] |
| Caulerpa spp.[1] | Padina[1] | Thalassia testudinum[1] |
| Cladium mariscus | Padina gymnospora[1] | Udotea flabellum[1] |
| Codium isthmocladum[1] | Penicillus capitatus[1] | Ulva sp.[1] |
| Digenea simplex[1] | Penicillus dumetosus[1] | Unidentified species |
| Distichlis spicata[2] | Penicillus spp.[1] | Yuzurua poiteaui[1] |

1 - Submerged Aquatic Vegetation, 2 - Coastal Wetlands

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